

Operational Amplifiers

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Outline

Basic Open-Loop Op-Amp Operation

Ideal vs. Real Op-Amp and Four the Rules

Impedance (Input, Output)

Negative Feedback Operation and Its Applications (Buffer, Inverting/Non-Inverting/Summing/Difference Amplifier, Integrator, Differentiator)

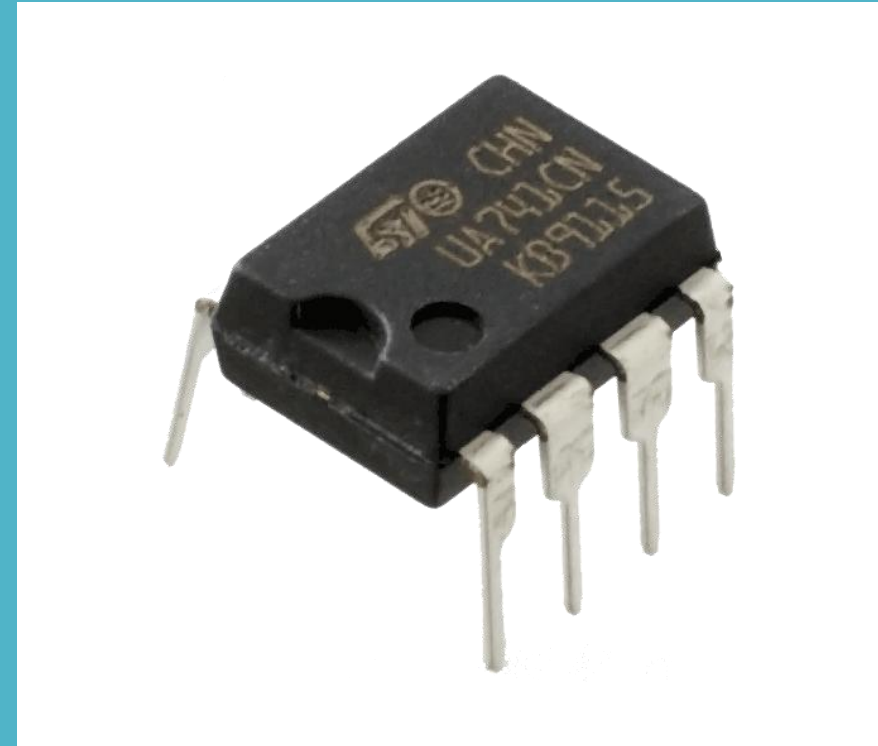
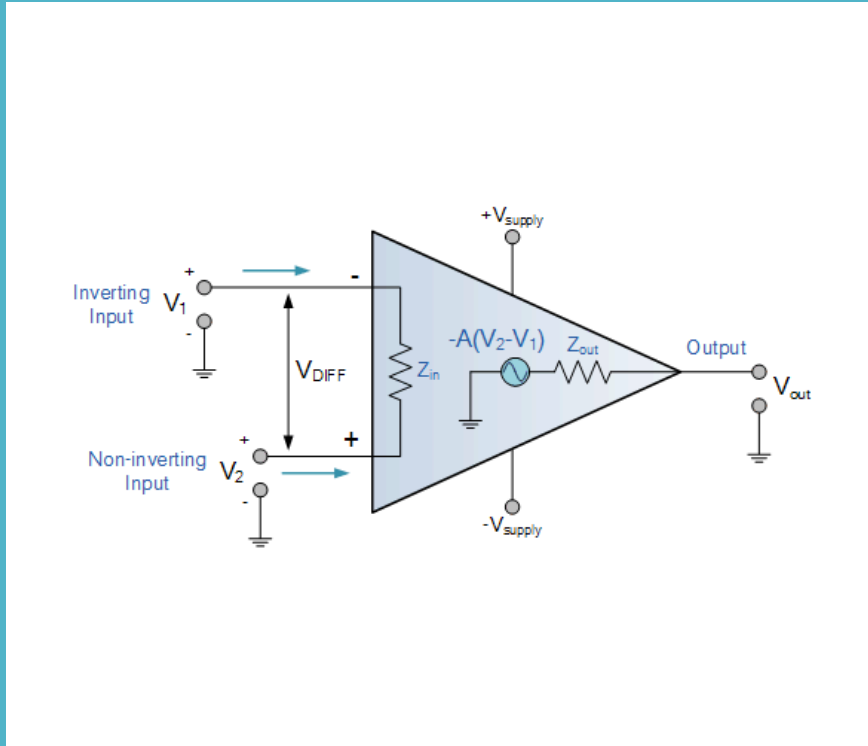
Negative Feedback Operation and Its Applications (Hysteresis, Comparator)

Types of Op-Amps (General-Purpose/Precision, Technologies, Programmable, Audio)

Op-Amp Power Supply (Single/Dual)

Op-Amp Specifications and Design Guidelines

Case Studies and Applications



Operational Amplifiers

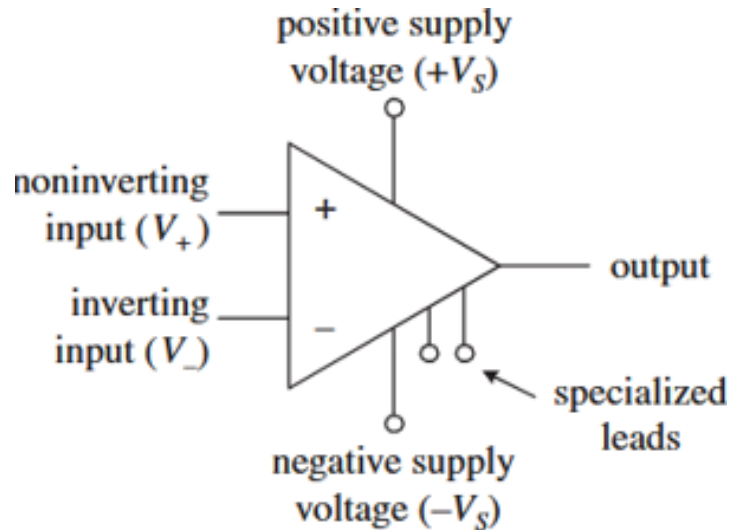
Operational Amplifiers (Op-Amps)

Op-Amps are high-performance **differential amplifiers** used in various electronic applications.

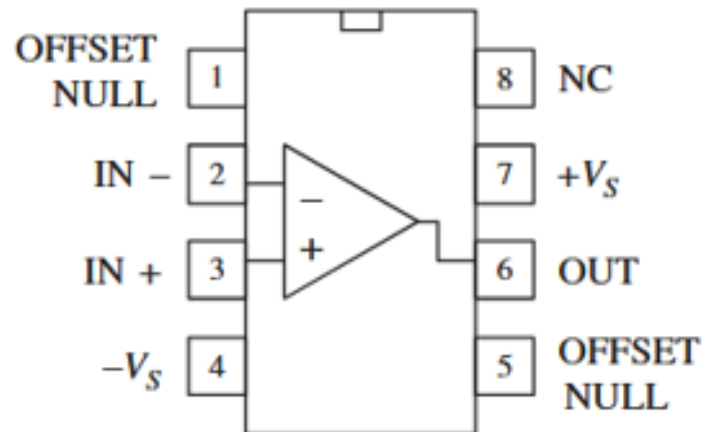
A typical op-amp has: **Noninverting (+)** and **Inverting (-)** inputs, **Output terminal**, **Dual DC power supplies**: $+V_s$ and $-V_s$, and (Optional) **Fine-tuning leads** (often omitted in diagrams).

Circuit schematics often **omit supply leads** — assume **dual supply** unless specified.

Though typically symmetric ($+V_s = -V_s$), **supply voltages can differ**, especially in **single-supply op-amps**.



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Differential Amplifier

A **differential amplifier** is an amplifier that **amplifies the difference between two input signals** (i.e., the difference between the **noninverting input (+)** and **inverting input (-)**), rather than amplifying a single input relative to ground.

This makes differential amplifiers ideal for rejecting **common-mode noise** (signals that appear on both inputs) and focusing only on the **true signal difference**.

$$V_{\text{out}} = A(V_+ - V_-)$$

A is the open-loop gain
(very large, e.g., 10^6)

Basic Op-Amp Operation: Open-Loop

If $V_- > V_+$, output drives to $-V_s$ (negative saturation).

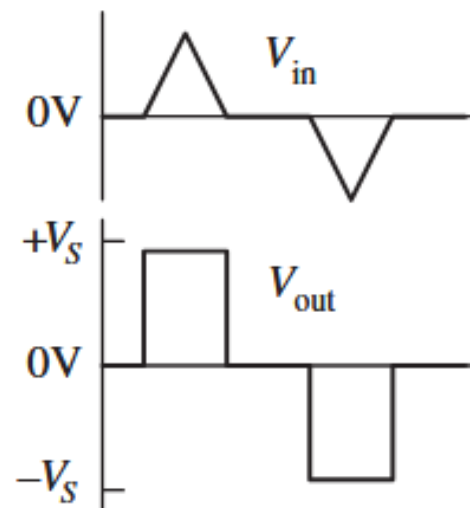
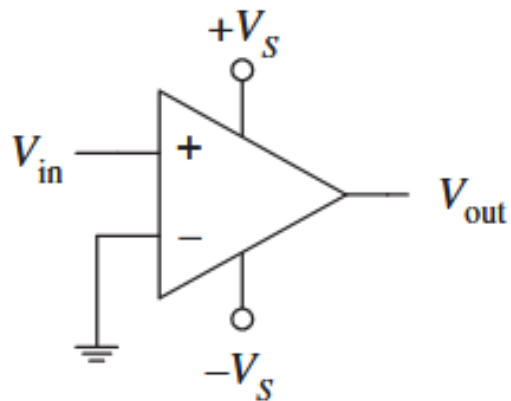
If $V_+ > V_-$, output drives to $+V_s$ (positive saturation).

Even a **tiny difference** between inputs causes the output to **saturate**.

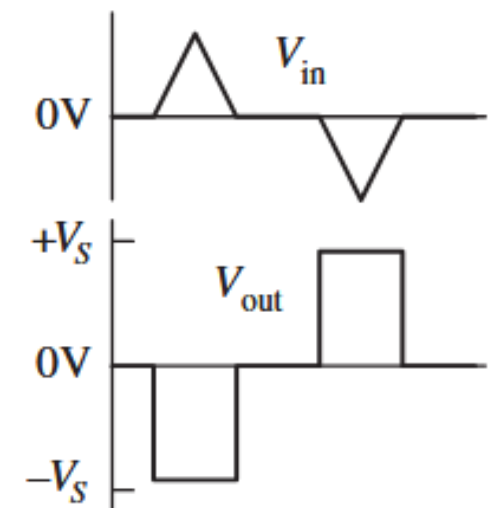
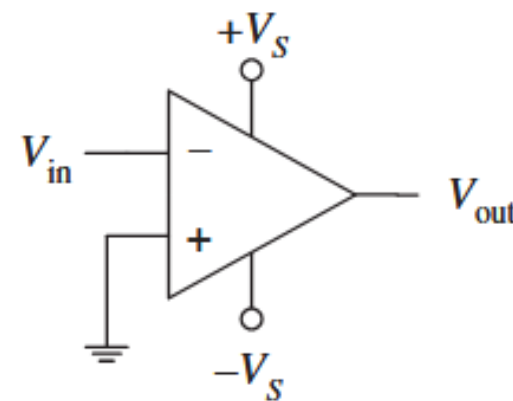
This behavior is known as **open-loop operation**.

Basic op-amp (open-loop) operation saturates output with even a small input difference — **limited practical use**.

Noninverting setup



Inverting setup



Negative Feedback Operation

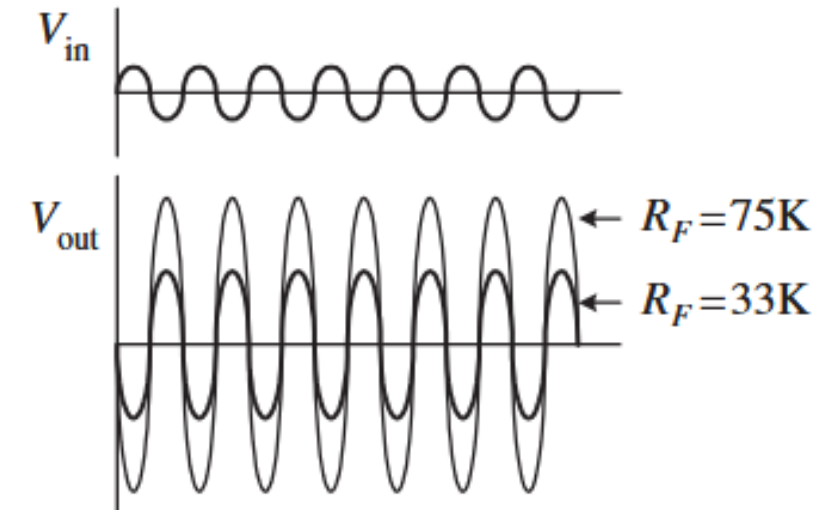
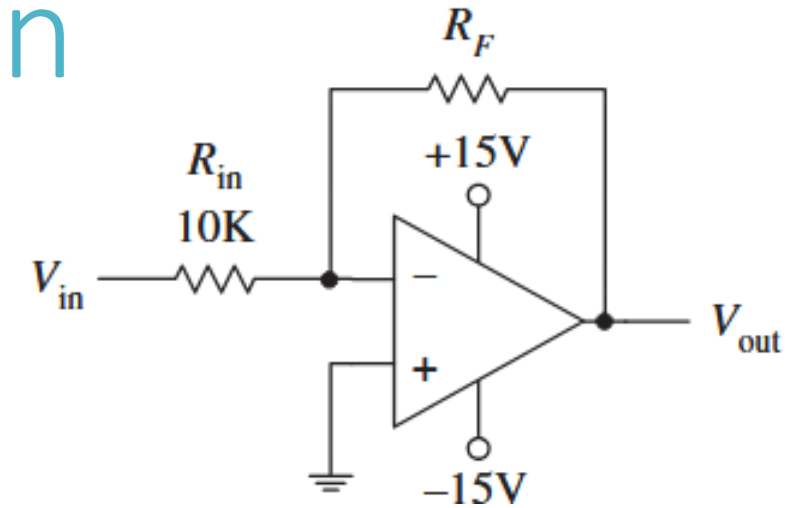
Negative feedback: feeds part of the output back to the inverting input.

Prevents saturation and allows for **controlled gain**.

$$V_{\text{out}} = -V_{\text{in}} \left(\frac{R_F}{R_{\text{in}}} \right)$$

Gain: $-R_F/R_{\text{in}}$ (output is inverted)

Gain increases with higher R_F , decreases with lower R_F



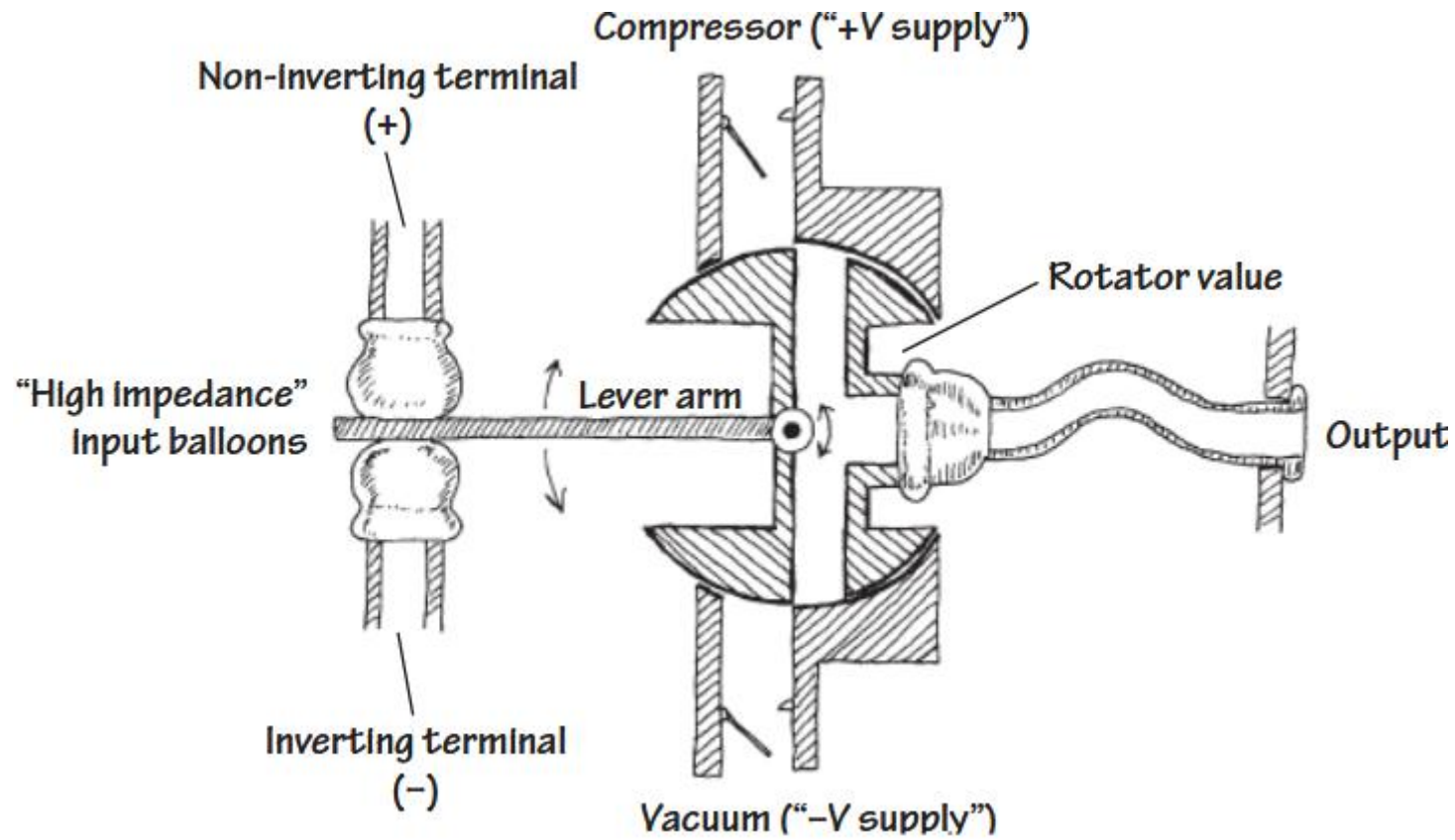
Applications of Negative Feedback

Amplification, regulation, conversion, filtering

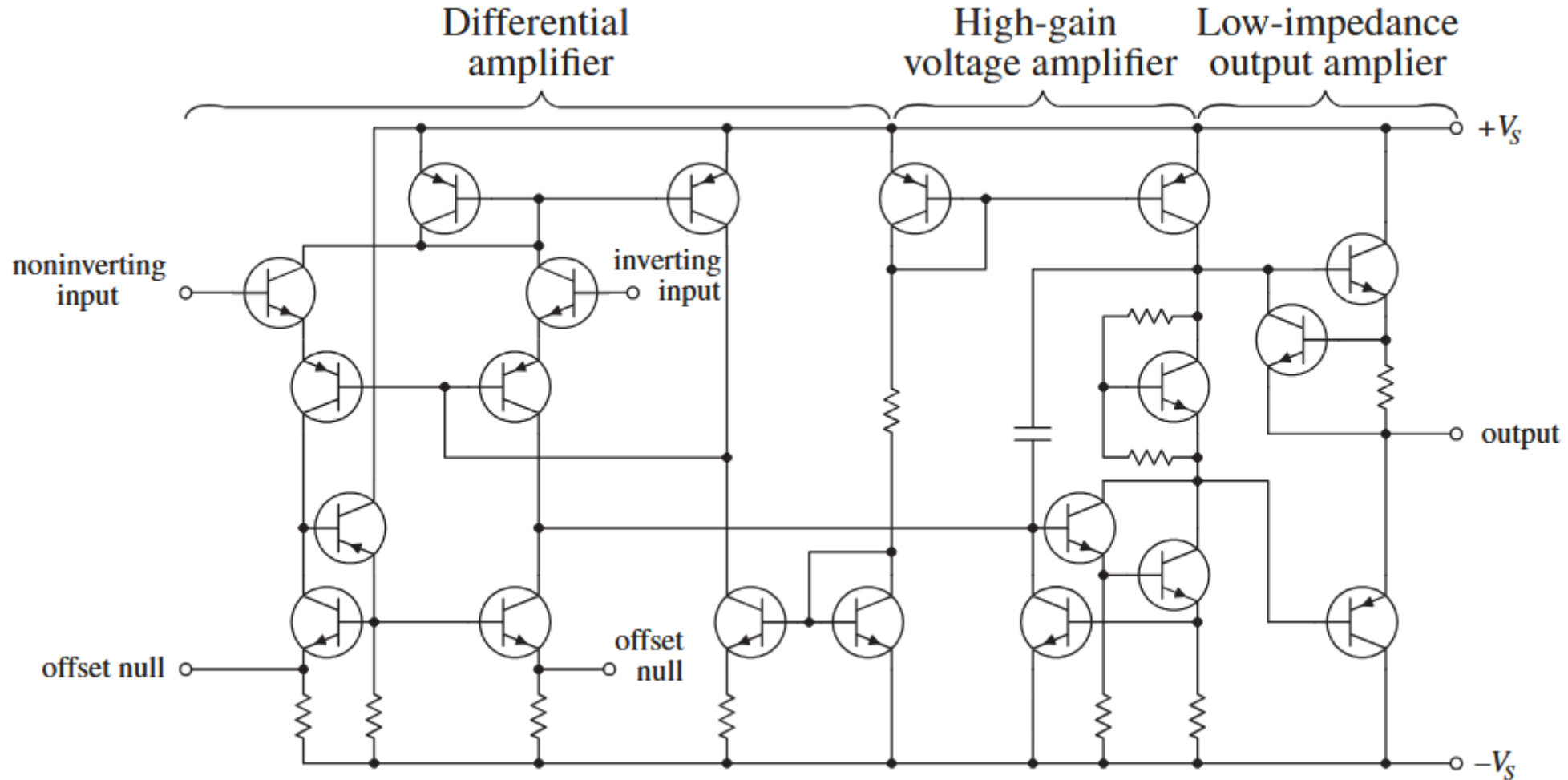
Math operations: add, subtract, integrate, differentiate

Waveform generation, rectification, detection, etc.

Op-Amp Water Analogy



Op-Amp Implementation



Darlington Transistor

A **Darlington transistor** (or **Darlington pair**) is a special arrangement of two **bipolar junction transistors (BJTs)** connected together to act as a single, high-gain amplifier.

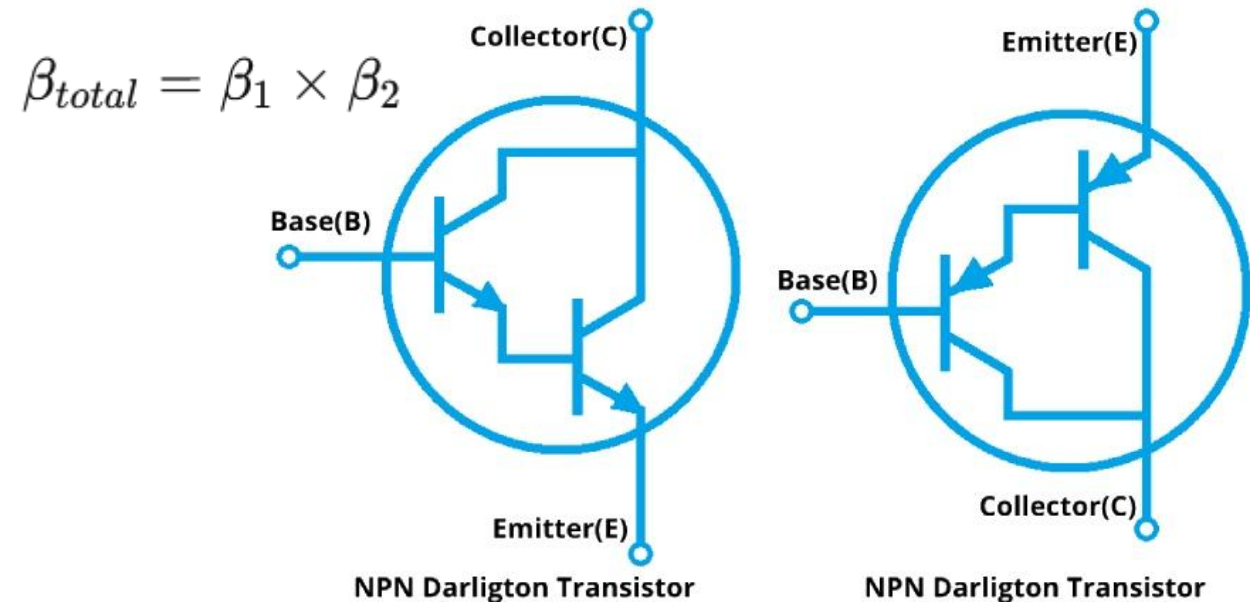
Key features:

- Very High Current Gain (β)

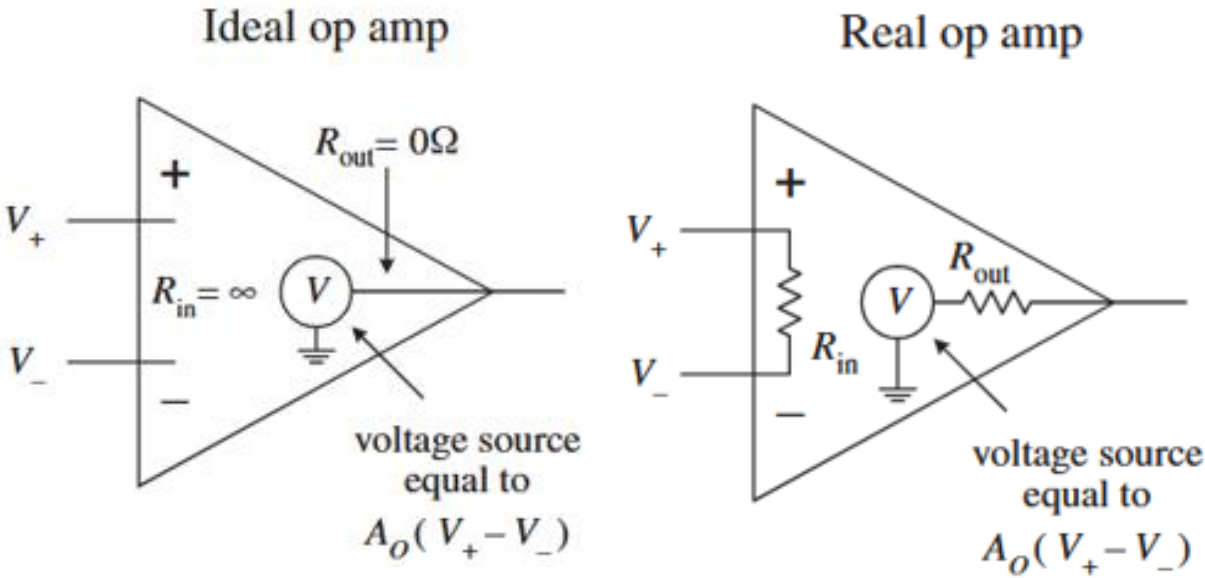
- High Input Impedance

- Low Output Impedance

- V_{BE} Voltage Drop (1.2V to 1.4V)



Ideal vs. Real Op-Amp



$$V_{out} = A_o(V_+ - V_-)$$

Parameter	Ideal Op-Amp	Real Op-Amp
Gain A_o	∞	10^4 to 10^6
Input Impedance R_{in}	∞	10^6 to $10^{12} \Omega$
Output Impedance R_{out}	0	10 to 1000 Ω
Input Current	0	Very small (pA to nA range)

Property	Ideal Value
Input impedance	Infinite
Output impedance	Zero
Open-loop gain (A_{OL})	Infinite
Bandwidth	Infinite
Offset voltage	Zero
Common-mode rejection ratio	Infinite

Input Impedance

$$Z_{in} = \frac{V_{in}}{I_{in}}$$

- V_{in} is the input voltage
- I_{in} is the current flowing into the input

Input impedance is the total impedance seen by a signal source at the input terminals of a device. It tells you **how much the device resists or loads the incoming signal**.

High Input Impedance:

Desirable for amplifiers and measurement devices.

Draws very little current from the signal source.

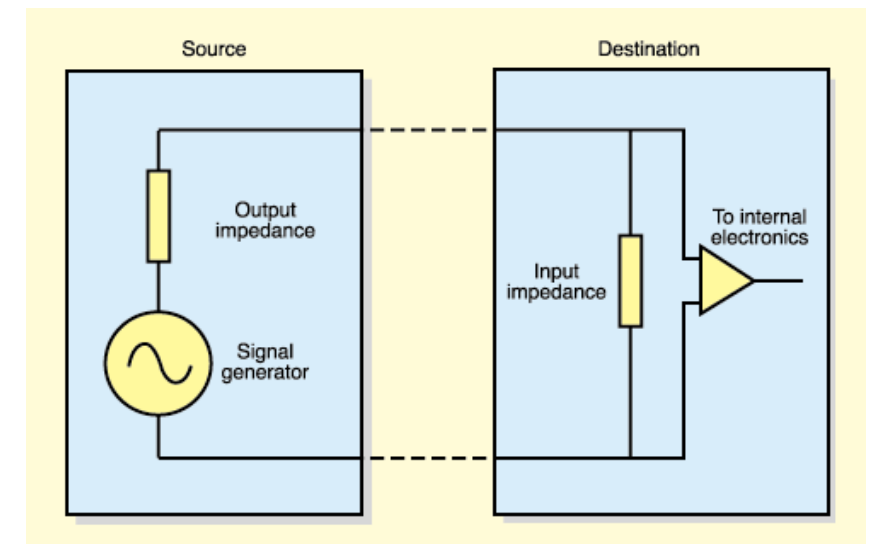
Prevents "loading" the source and keeps the original signal intact.

Examples:

Op-amp (ideal): infinite input impedance.

Multimeter: ~10 MΩ input impedance.

Oscilloscope probe: typically, 1 MΩ or more.



Output Impedance

$$Z_{out} = \frac{V_{open} - V_{load}}{I_{load}}$$

V_{open} is the output voltage with no load.

V_{load} is the output voltage with a load connected.

I_{load} is the current into the load.

Output impedance is the impedance "seen" by the load connected to the output of a device. It reflects **how well the device can drive a load** without the output voltage dropping due to internal resistance.

Low Output Impedance:

Desirable in amplifiers and buffers.

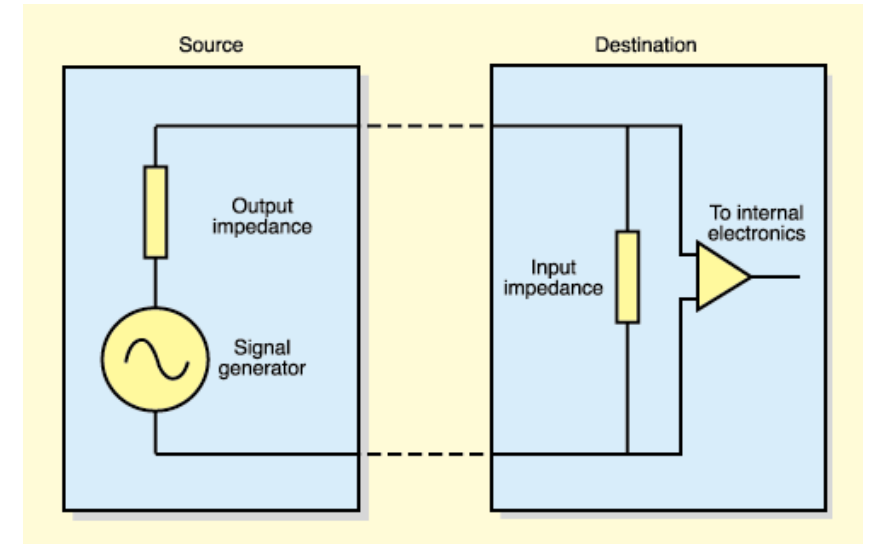
Delivers more power to the load.

Keeps output voltage stable regardless of load.

Examples:

Voltage source (ideal): 0 Ω output impedance.

Op-amp (ideal): 0 Ω ; real op-amps: typically <100 Ω .



Impedance Matching

In **audio** or **RF** circuits, **impedance matching** is critical for **maximum power transfer**.

In **signal processing**, you often want **high input impedance** (to not load the source) and **low output impedance** (to drive loads efficiently).

Op-Amp Rules

Rule 1: For an ideal op amp, the open-loop voltage gain is infinite ($A_o = \infty$). For a real op amp, the gain is a finite value, typically between 10^4 to 10^6 .

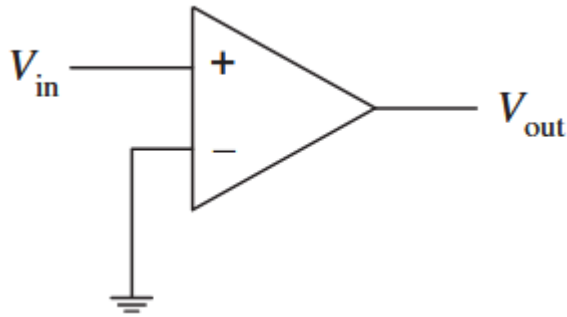
Rule 2: For an ideal op amp, the input impedance is infinite ($R_{in} = \infty$). For a real op amp, the input impedance is finite, typically between 10^6 (e.g., typical bipolar op amp) to $10^{12} \Omega$ (e.g., typical JFET op amp). The output impedance for an ideal op amp is zero ($R_{out} = 0$). For a real op amp, R_{out} is typically between 10 to 1000 Ω .

Rule 3: The input terminals of an ideal op amp draw no current. Practically speaking, this is true for a real op amp as well—the actual amount of input current is usually (but not always) insignificantly small, typically within the picoamps (e.g., typical JFET op amp) to nanoamps (e.g., typical bipolar op amp) range.

Rule 4: Whenever an op amp senses a voltage difference between its inverting and noninverting inputs, it responds by feeding back as much current/voltage through the feedback network as is necessary to keep this difference equal to zero ($V_+ - V_- = 0$). This rule only applies for negative feedback.

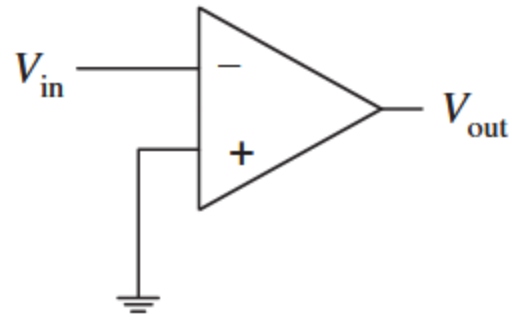
Examples

Solve for the gain ($V_{\text{out}}/V_{\text{in}}$) of the circuit below.



$$\begin{aligned} V_{\text{out}} &= A_o(V_+ - V_-) \\ &= A_o(V_{\text{in}} - 0 \text{ V}) = A_o V_{\text{in}} \end{aligned}$$

$$\text{Gain} = \frac{V_{\text{out}}}{V_{\text{in}}} = A_o$$



$$\begin{aligned} V_{\text{out}} &= A_o(V_+ - V_-) \\ &= A_o(0 \text{ V} - V_{\text{in}}) = -A_o V_{\text{in}} \end{aligned}$$

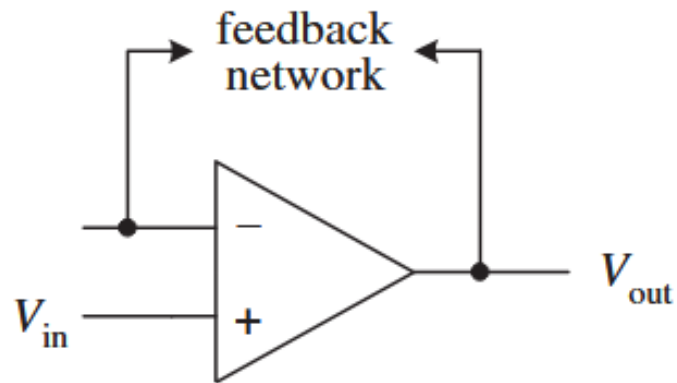
$$\text{Gain} = \frac{V_{\text{out}}}{V_{\text{in}}} = -A_o$$

Negative Feedback Cont.

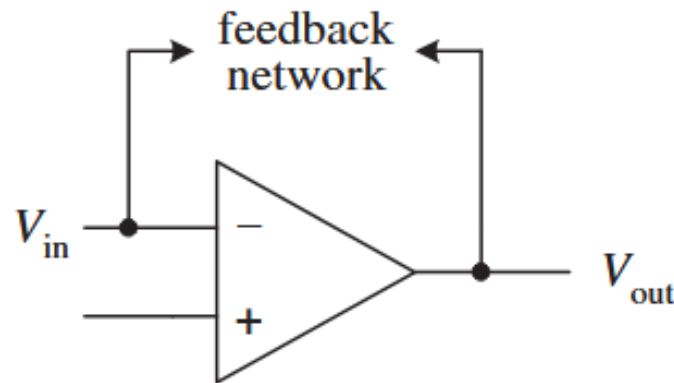
Negative feedback occurs when part of the **output voltage** is **fed back** to the **inverting input** of an op-amp. It stabilizes gain and improves linearity. Feedback can be through a **resistor, capacitor, complex circuit**, or simply a **wire**.

There are two basic kinds of negative feedback, **voltage feedback** and **operational feedback**.

Voltage Feedback



Operational Feedback



$$V_- = fV_{out}$$

$$V_{out} = A_0(V_+ - fV_{out})$$

Where f is a fraction of the voltage “sent” back from **Vout**

Negative Feedback Cont.

The feedback fraction f helps conceptually but is **not required** in calculations.

For an **ideal op-amp** ($A_o \rightarrow \infty$):

Therefore, we can say that **in a negative feedback configuration**, the op-amp **adjusts the output** to make $V_+ = V_-$.

This rule is **essential** for solving op-amp circuits easily and accurately.

$$V_- = fV_{\text{out}}$$

$$V_{\text{out}} = A_o(V_+ - fV_{\text{out}})$$

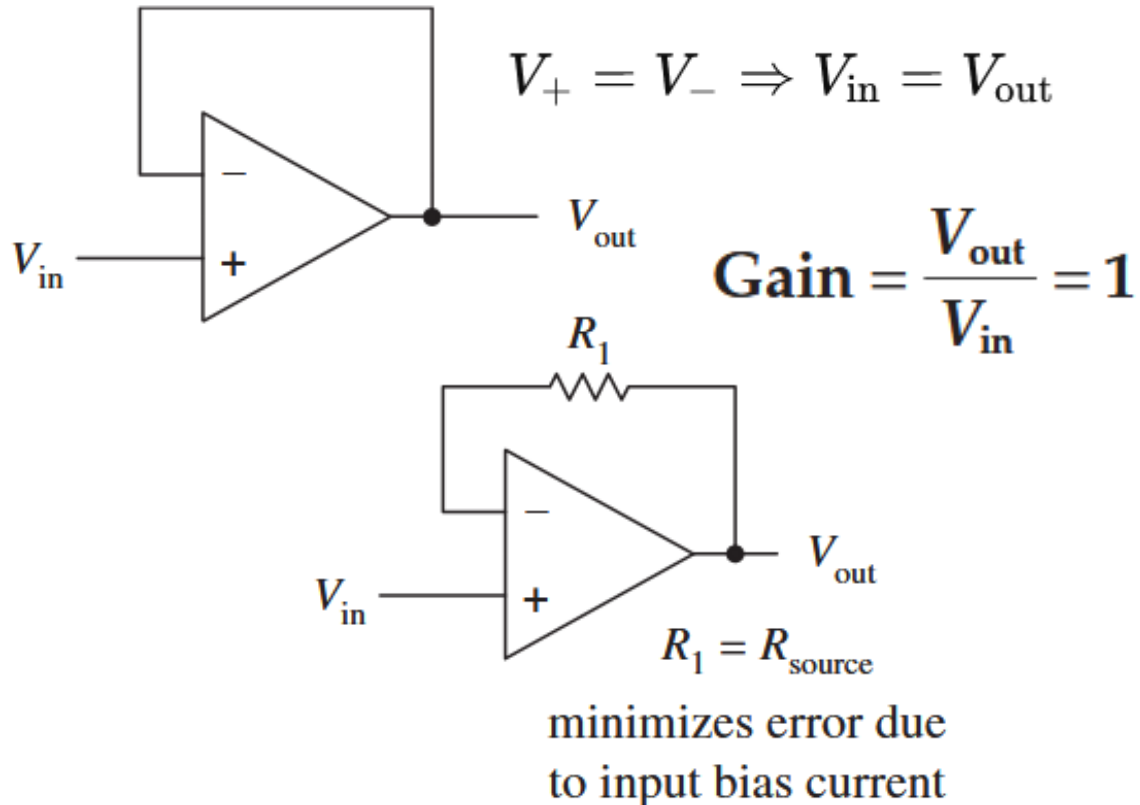
$$V_{\text{out}} = A_o(V_+ - V_-)$$

$$\frac{V_{\text{out}}}{A_o} = V_+ - V_- \Rightarrow V_+ - V_- = 0$$

Negative Feedback Example Problems:

1. Buffer (Unity Gain Amplifier)

Solve for the gain ($V_{\text{out}}/V_{\text{in}}$) of the circuit below.

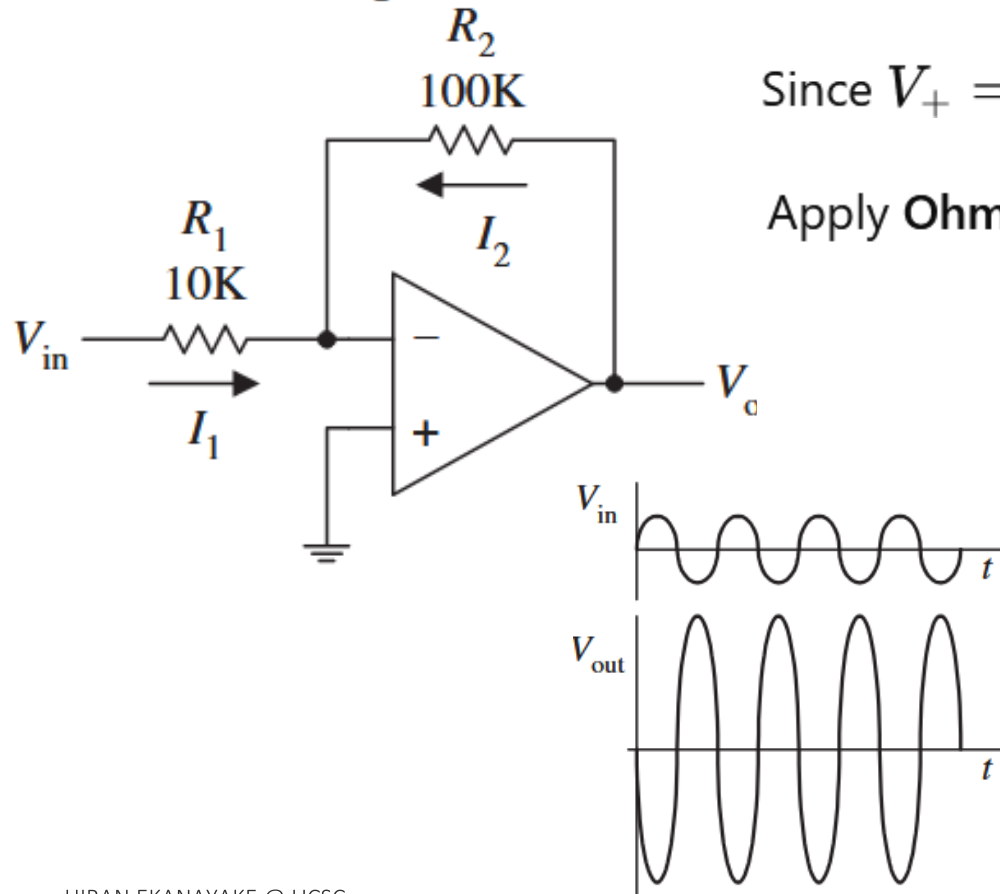


- **Purpose:**
Acts as a **buffer** (not a signal amplifier)
 - Very high input impedance
 - Very low output impedance
→ Ideal for **isolating stages** in a circuit
- **Practical Note:**
A feedback resistor may be added to reduce **voltage offset errors** from input bias currents.
→ Set feedback resistor $\approx$ source resistance

Negative Feedback Example Problems:

2. Inverting Amplifier

Solve for the gain ($V_{\text{out}}/V_{\text{in}}$) of the circuit below.



Since $V_+ = 0$ (grounded), $V_- = 0$ as well.

Apply **Ohm's Law**:

$$I_1 = \frac{V_{\text{in}}}{R_1}, \quad I_2 = \frac{V_{\text{out}}}{R_2}$$

Apply **Kirchhoff's current law**:

$$I_1 + I_2 = 0 \Rightarrow I_2 = -I_1$$

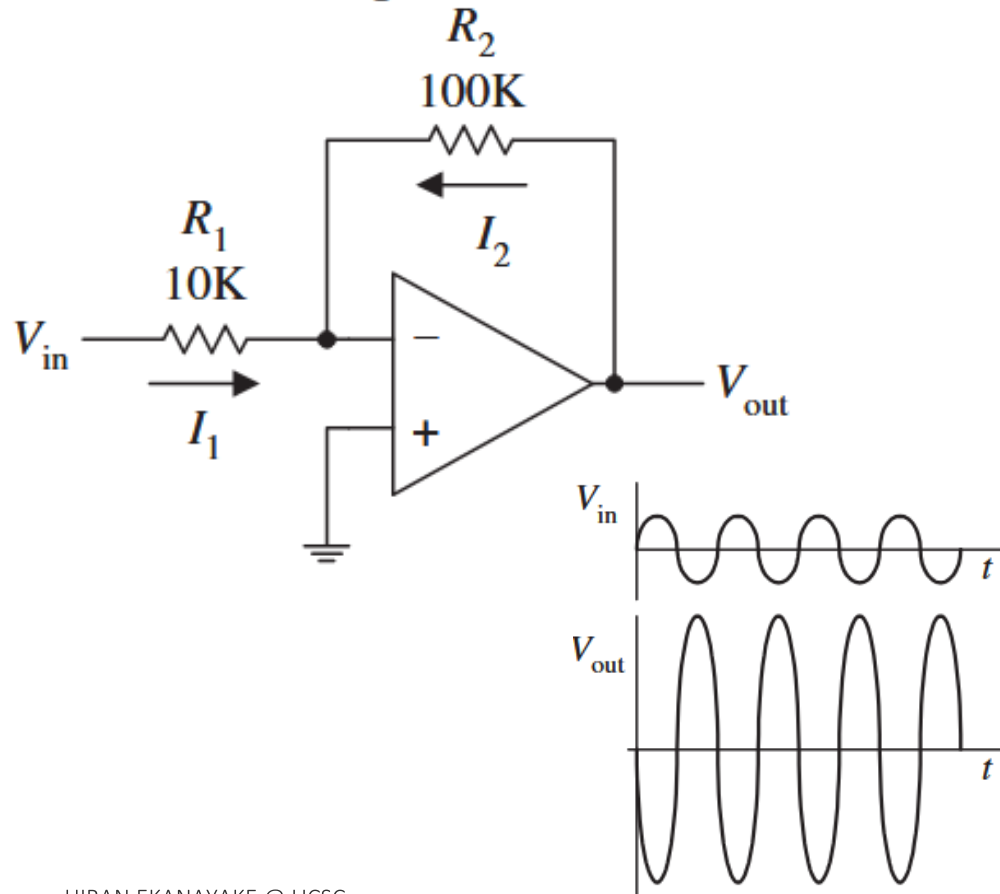
$$\frac{V_{\text{out}}}{R_2} = -\frac{V_{\text{in}}}{R_1} \Rightarrow V_{\text{out}} = -\left(\frac{R_2}{R_1}\right) V_{\text{in}}$$

Because an ideal op amp has infinite input impedance, no current will enter its inverting terminal.

Negative Feedback Example Problems:

2. Inverting Amplifier Cont.

Solve for the gain ($V_{\text{out}}/V_{\text{in}}$) of the circuit below.



$$\frac{V_{\text{out}}}{R_2} = -\frac{V_{\text{in}}}{R_1} \Rightarrow V_{\text{out}} = -\left(\frac{R_2}{R_1}\right) V_{\text{in}}$$

Negative sign $\rightarrow$ 180° phase shift (inversion)

If $R_1 = R_2$, gain = -1 $\rightarrow$ Unity-gain inverter

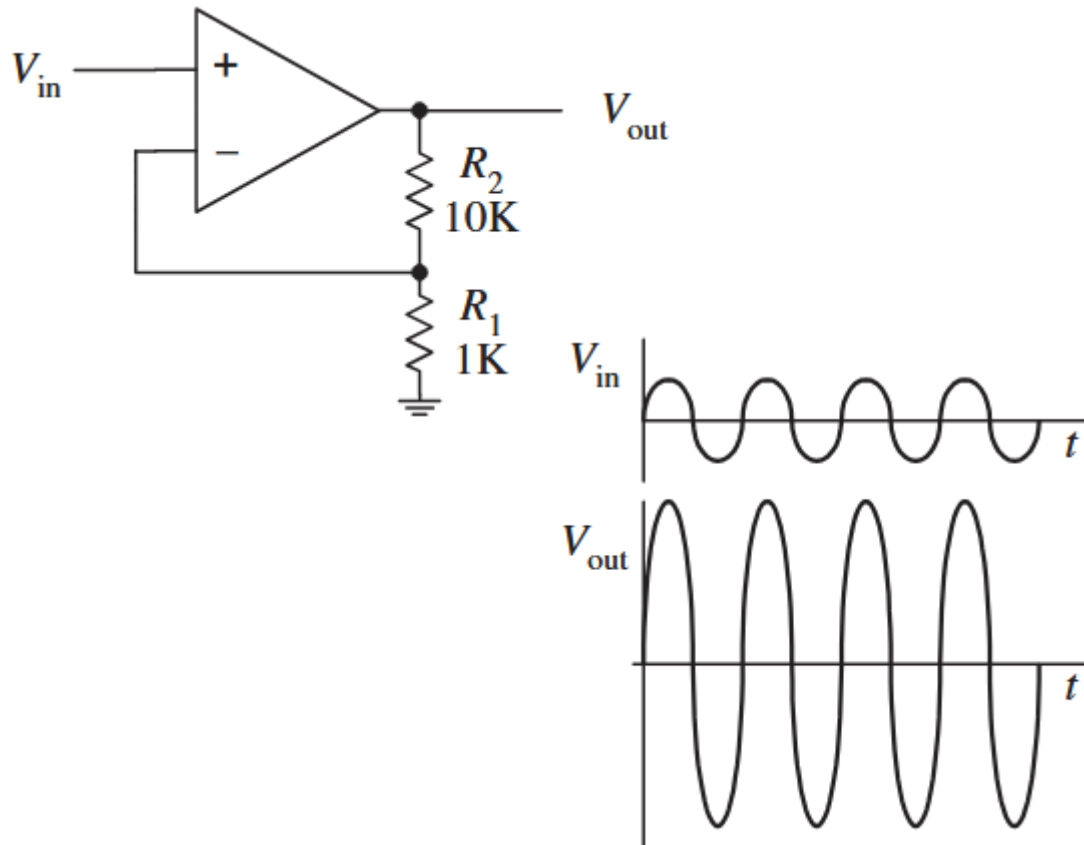
Practical Note:

For real op amps with input bias currents, add a resistor $R = R_1 || R_2$ between the noninverting input and ground to minimize offset voltage errors.

Negative Feedback Example Problems:

3. Noninverting Amplifier

Solve for the gain ($V_{\text{out}}/V_{\text{in}}$) of the circuit below.



$$V_- = \frac{R_1}{R_1 + R_2} V_{\text{out}} = V_{\text{in}}$$

$$\text{Gain} = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{R_1 + R_2}{R_1} = 1 + \frac{R_2}{R_1}$$

Output is **in phase** with input → **noninverting**

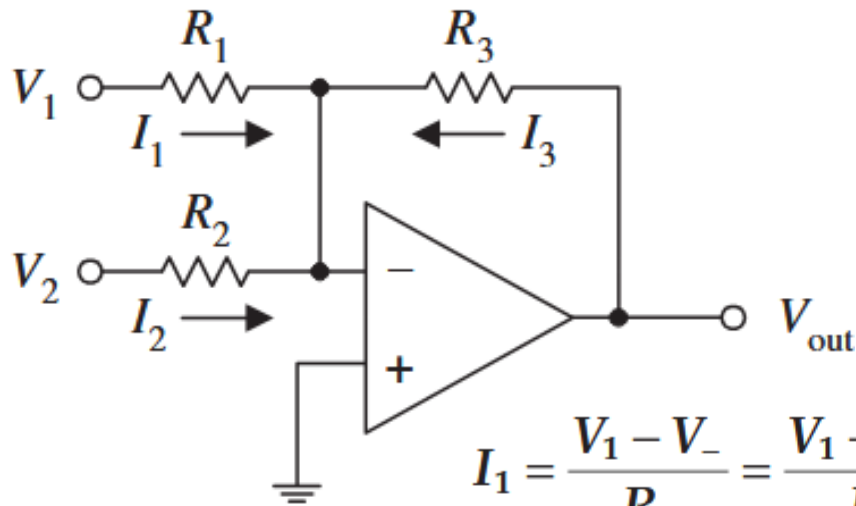
To reduce **voltage offset errors** from input bias current, use:

$$R = R_1 || R_2 = R_{\text{source}}$$

Negative Feedback Example Problems:

4. Summing Amplifier (Inverting)

Solve for the gain ($V_{\text{out}}/V_{\text{in}}$) of the circuit below.



$$I_1 = \frac{V_1 - V_-}{R_1} = \frac{V_1 - 0\text{ V}}{R_1} = \frac{V_1}{R_1}$$

$$I_2 = \frac{V_2 - V_-}{R_2} = \frac{V_2 - 0\text{ V}}{R_2} = \frac{V_2}{R_2}$$

$$I_3 = \frac{V_{\text{out}} - V_-}{R_3} = \frac{V_{\text{out}} - 0\text{ V}}{R_3} = \frac{V_{\text{out}}}{R_3}$$

$$I_3 = -(I_1 + I_2) \Rightarrow \frac{V_{\text{out}}}{R_3} = -\left(\frac{V_1}{R_1} + \frac{V_2}{R_2}\right)$$

$$V_{\text{out}} = -R_3 \left(\frac{V_1}{R_1} + \frac{V_2}{R_2}\right)$$

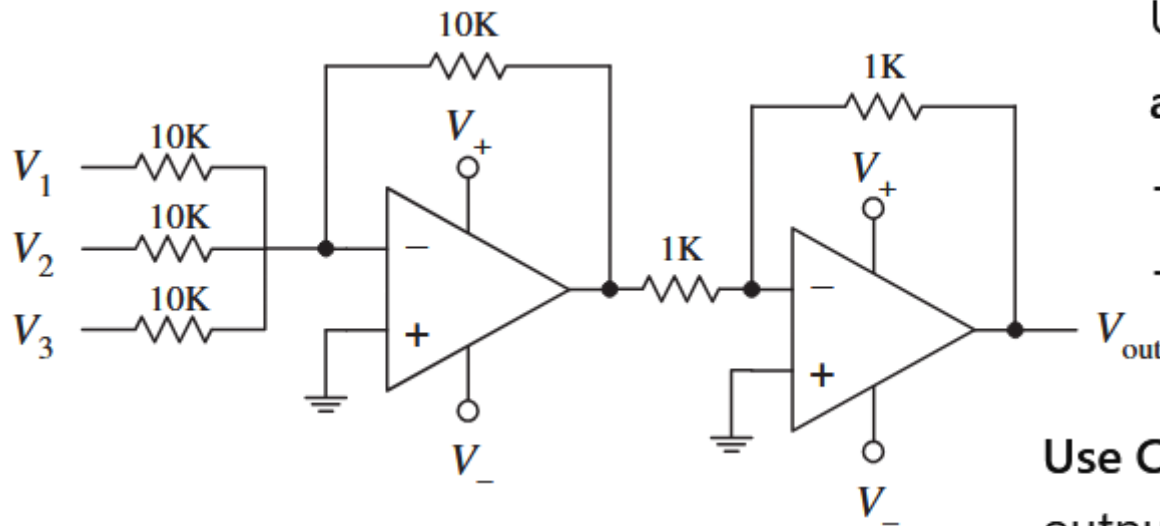
If $R_1 = R_2 = R_3 = R$, then

$$V_{\text{out}} = -(V_1 + V_2)$$

Negative Feedback Example Problems:

4. Summing Amplifier (Noninverting)

Solve for the gain ($V_{\text{out}}/V_{\text{in}}$) of the circuit below.



Add an **inverting stage** to get:

$$V_{\text{out}} = V_1 + V_2 + V_3$$

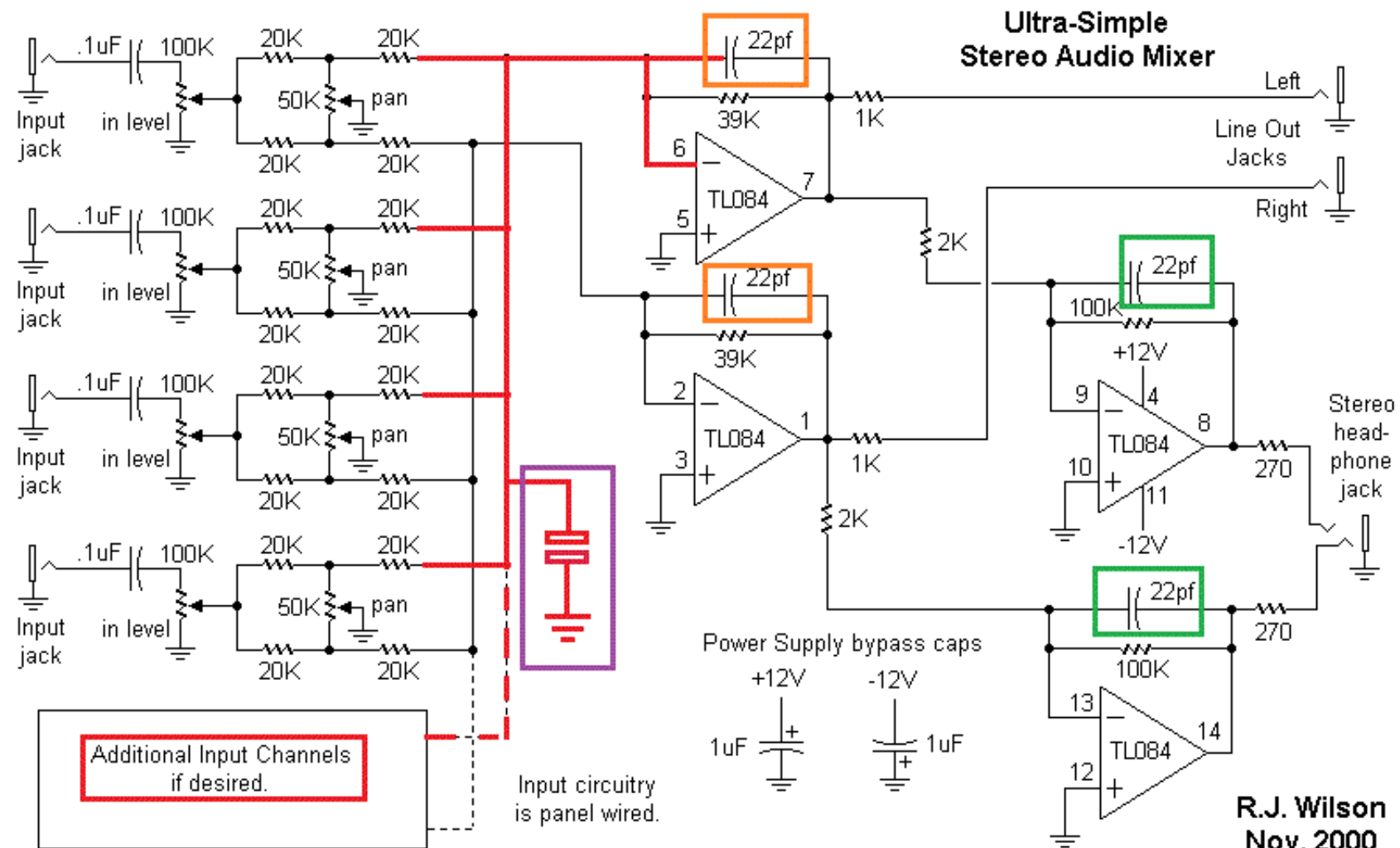
Use a compensation resistor between the **noninverting input and ground**

→ Value = **parallel of all input resistors**

→ Minimizes **offset error** from input bias currents

Use Case: Combines multiple input signals into a single weighted output—common in audio mixing, signal processing, and DAC circuits.

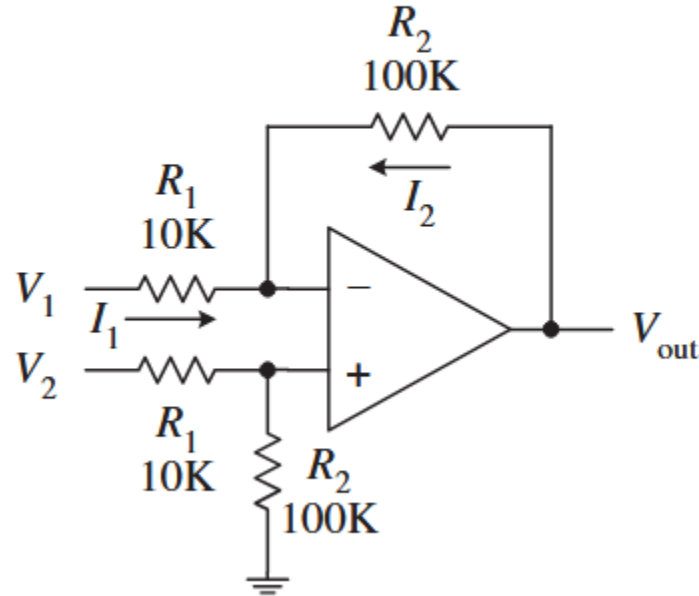
Audio Mixer



Negative Feedback Example Problems:

5. Difference Amplifier

Determine V_{out} .



$$V_+ = \frac{R_2}{R_1 + R_2} V_2$$

$$(I_1 = I_2):$$

$$\frac{V_1 - V_-}{R_1} = \frac{V_- - V_{\text{out}}}{R_2}$$

$$V_+ = V_-$$

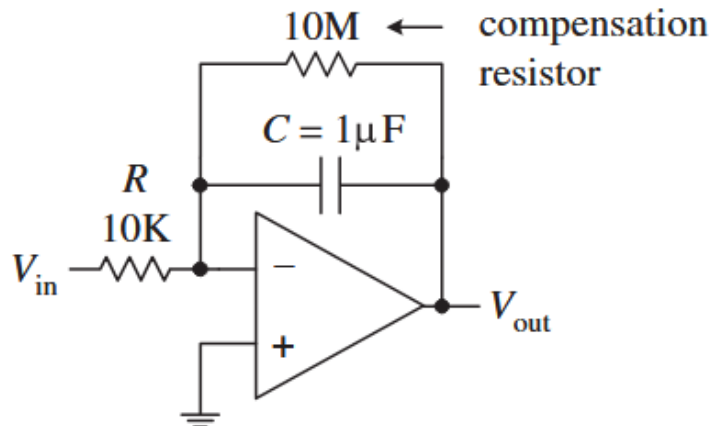
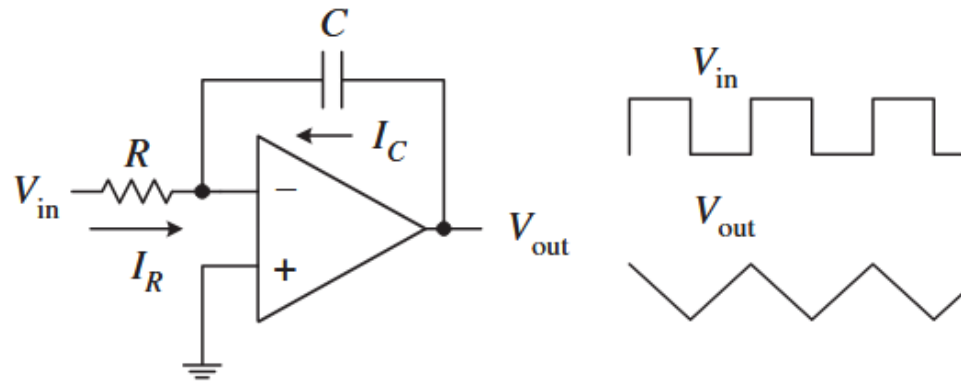
$$V_{\text{out}} = \frac{R_2}{R_1} (V_2 - V_1)$$

If you set $R_1 = R_2$, then $V_{\text{out}} = V_2 - V_1$.

Negative Feedback Example Problems:

6. Integrator

Solve for V_{out} in terms of V_{in} .



$$I_R = \frac{V_{in} - V_-}{R} = \frac{V_{in} - 0V}{R} = \frac{V_{in}}{R}$$

$$I_C = C \frac{dV}{dt} = C \frac{d(V_{out} - V_-)}{dt} = C \frac{d(V_{out} - 0V)}{dt} = C \frac{dV_{out}}{dt}$$

$$I_R + I_C = 0$$

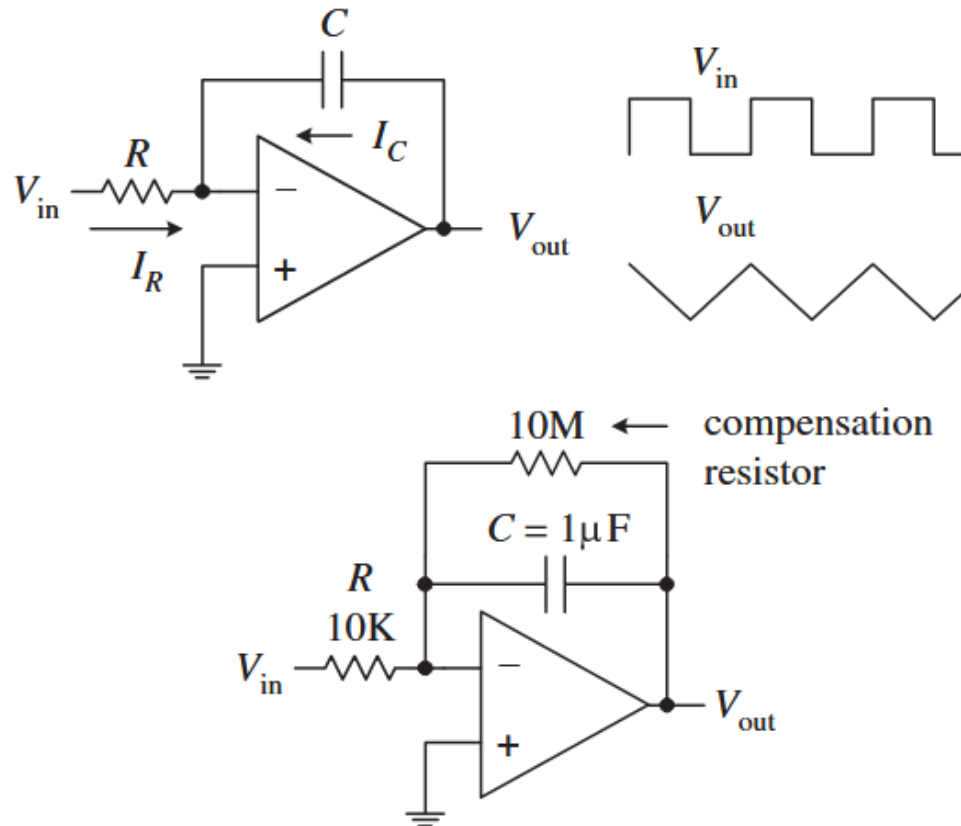
$$dV_{out} = -\frac{1}{RC} V_{in} dt$$

$$V_{out}(t) = -\frac{1}{RC} \int V_{in}(t) dt$$

Negative Feedback Example Problems:

6. Integrator Cont.

Solve for V_{out} in terms of V_{in} .



Drift and saturation due to op-amp non-idealities (offset voltage, input bias current)

Fix: Add a large resistor in parallel with the capacitor to allow **DC feedback**

Compensation resistor (connected from noninverting input to ground):

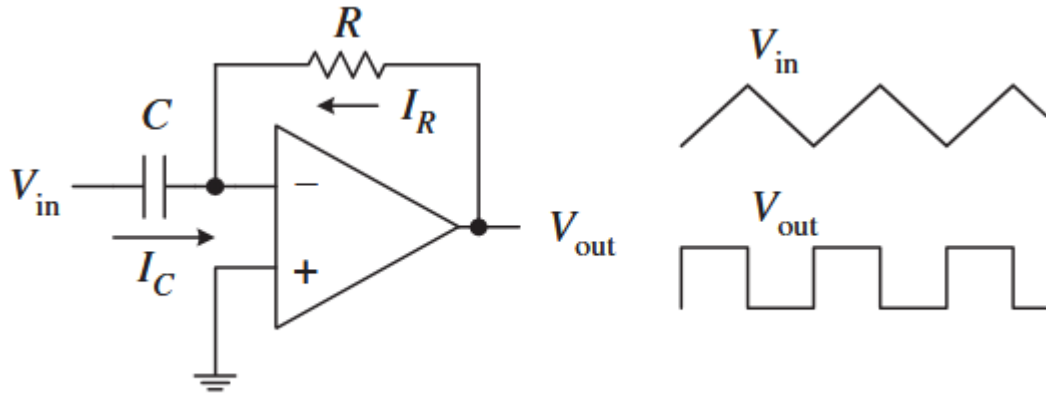
Value = parallel of **input resistor** and **feedback resistor**

→ Minimizes **offset error**

Use Case: Signal integration in analog computing, waveform generation, and filters.

Negative Feedback Example Problems:

7. Differentiator



$$I_C = C \frac{dV}{dt} = C \frac{d(V_{in} - V_-)}{dt} = C \frac{d(V_{in} - 0V)}{dt} = C \frac{dV_{in}}{dt}$$

$$I_R = \frac{V_{out} - V_-}{R} = \frac{V_{out} - 0V}{R} = \frac{V_{out}}{R}$$

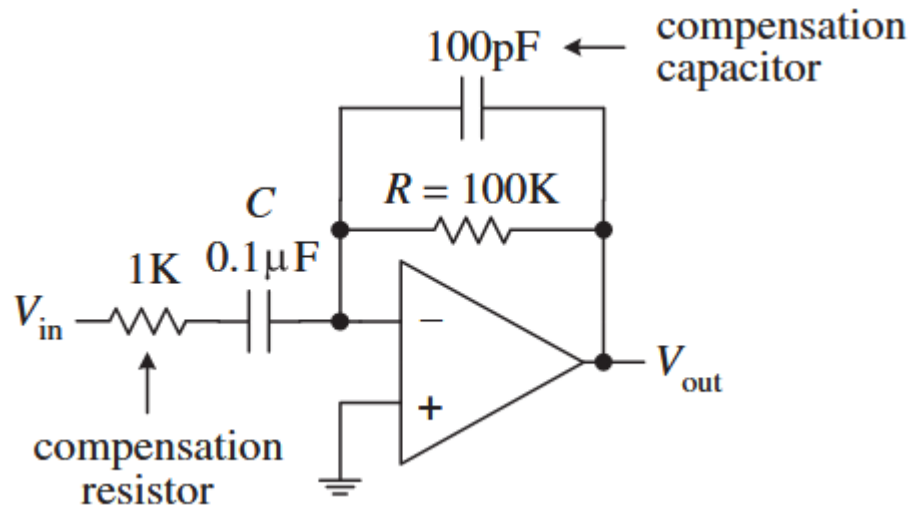
$$I_R + I_C = 0 \Rightarrow V_{out}(t) = -RC \frac{dV_{in}(t)}{dt}$$

Output is the **derivative** of the input signal

High AC gain $\rightarrow$ very **sensitive to noise**

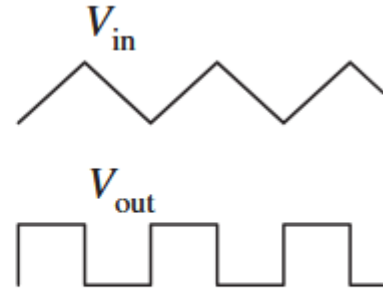
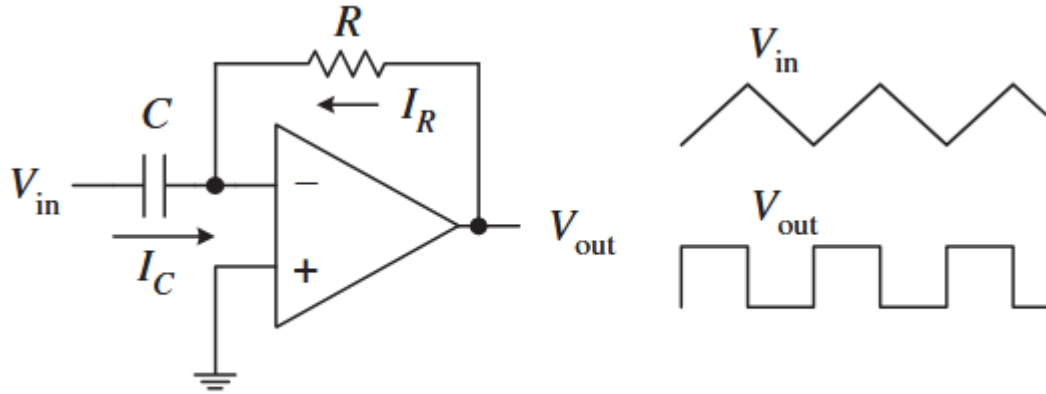
Phase lag from RC network $\rightarrow$ **stability issues**

Add a **feedback capacitor** and **series input resistor**



Negative Feedback Example Problems:

7. Differentiator Cont.

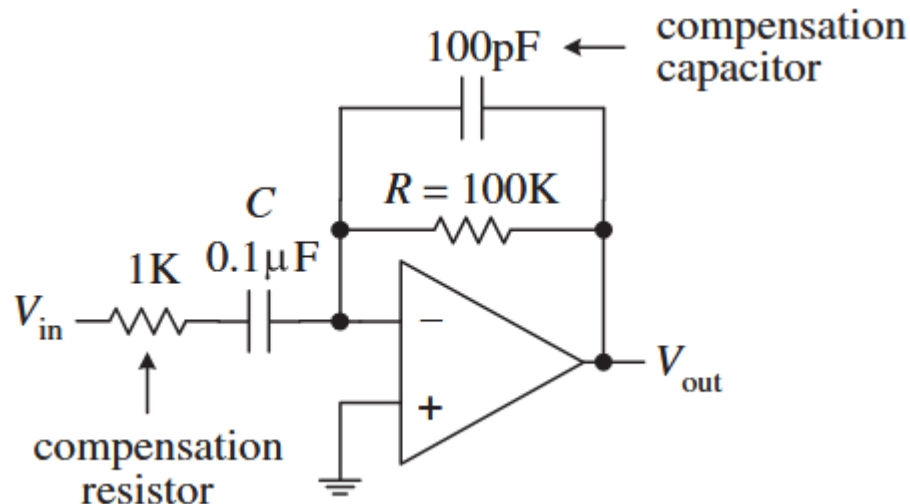


Add a **feedback capacitor** and **series input resistor**

Provide **high-frequency roll-off** (reduces noise)

Introduce **90° phase lead** (improves stability)

Limit usable **frequency range** (at high frequencies, circuit behaves like an integrator)



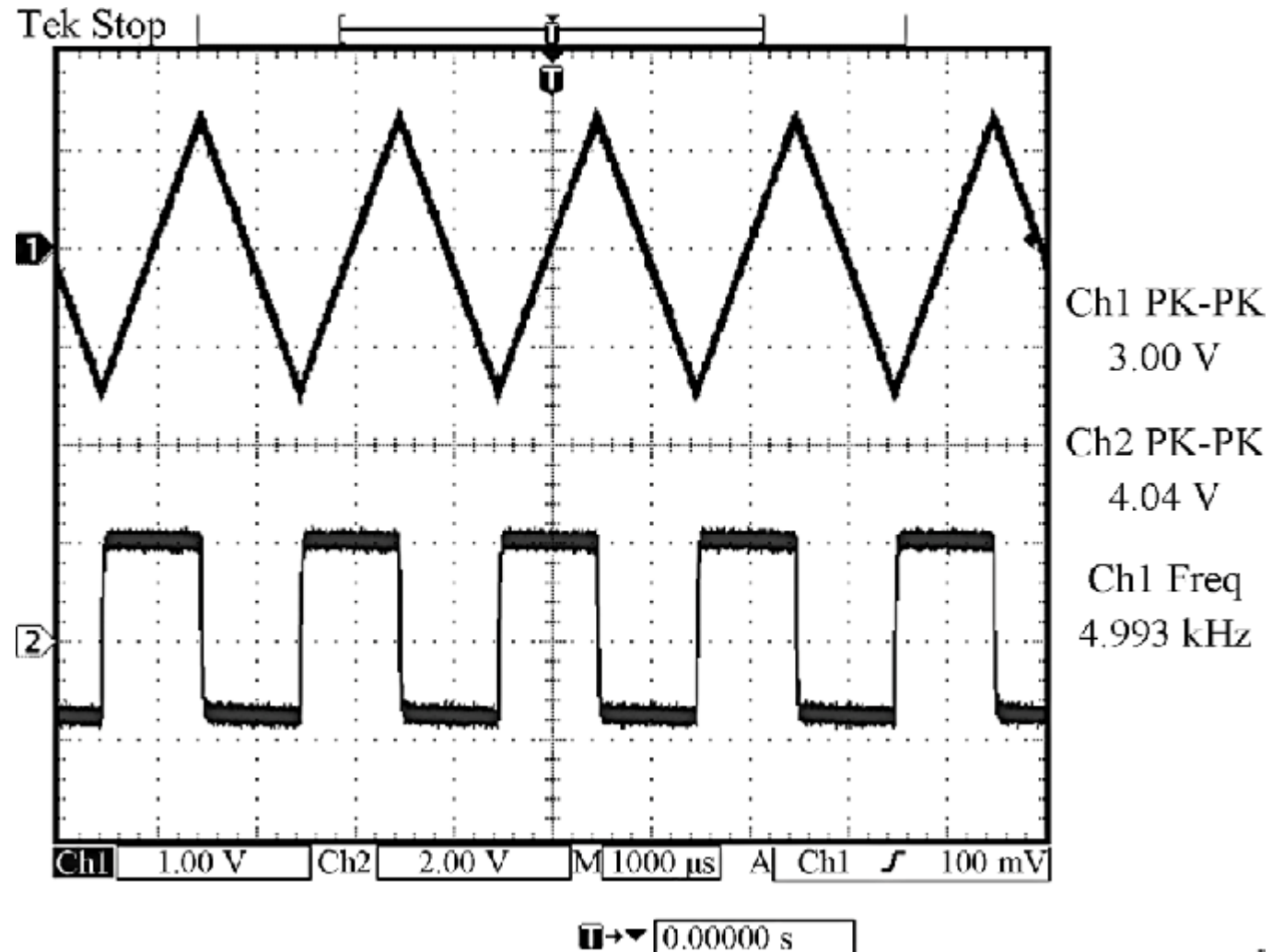
Add a resistor from **noninverting input to ground**

Value = **feedback resistor**

→ Minimizes **offset error** from input bias current

Use Case: Signal differentiation in edge detection, waveform shaping, and analog computing—best used with bandwidth and noise control.

Voltage Differentiator



Positive Feedback

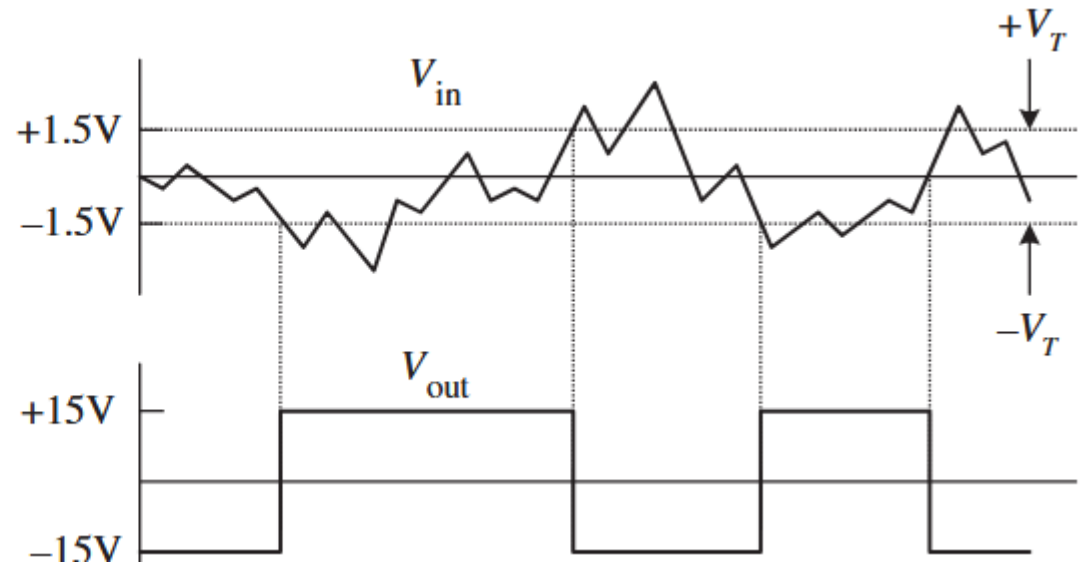
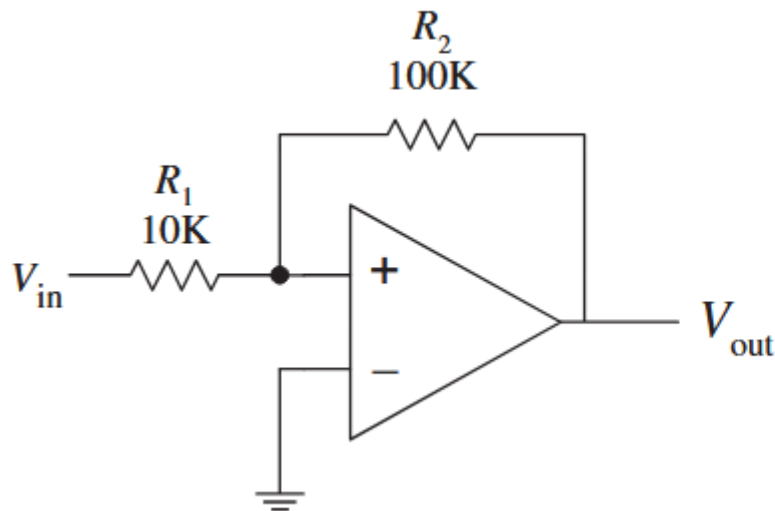
Positive feedback occurs when a portion of the output voltage is fed back to the **non-inverting input** of the op-amp.

It reinforces the direction of the output, driving it further toward saturation:

$V_{out} = A_o(fV_{out} - V_-)$. Here, fV_{out} increases the output, unlike **negative feedback** where it subtracts.

Negative feedback → Stabilizes gain, prevents saturation (desirable).

Positive feedback → Drives output to extremes (generally undesirable), but useful in specific cases.



Use of Positive Feedback in Comparators

Application: Positive feedback is beneficial in comparator circuits to introduce **hysteresis** and reduce noise sensitivity.

Hysteresis: Adds **two threshold voltages** ($\pm V_T$), making the output less sensitive to small input changes or noise.

Hysteresis Example

Given: Output saturation = ± 15 V, Resistor ratio $R_2/R_1 = 10$

Negative threshold:

$$-V_T = \frac{-V_{\text{out}}}{R_2/R_1} = \frac{-15}{10} = -1.5 \text{ V}$$

Positive threshold:

$$+V_T = \frac{+15}{10} = +1.5 \text{ V}$$

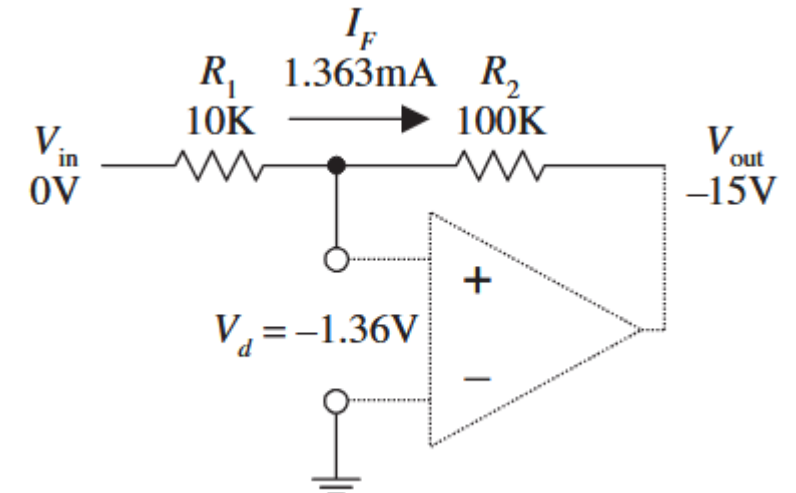
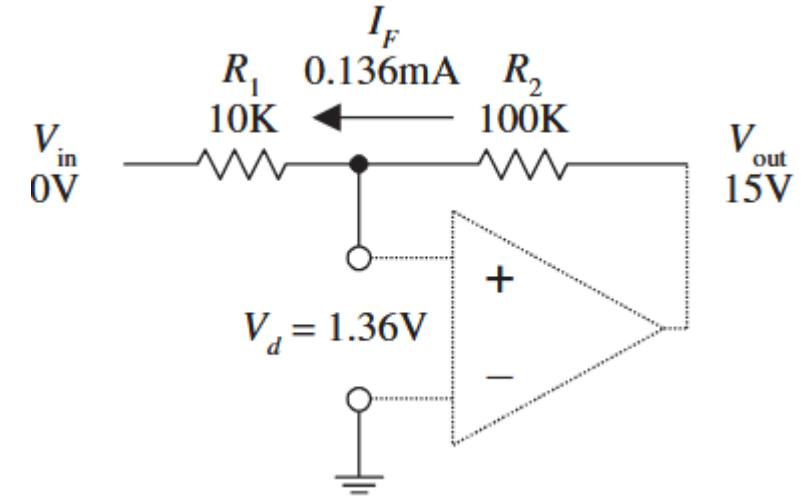
Hysteresis Voltage:

$$V_h = +V_T - (-V_T) = 3 \text{ V}$$

Threshold voltage = at which time the output switches states (i.e., when $V_d=0$ of the rising or falling edge of V_d).

$$I_F = (V_{\text{out}} - V_{\text{in}}) / (R_1 + R_2)$$

$$V_d = I_F R_1$$



Hysteresis Example Cont.

Given: Output saturation = ± 15 V, Resistor ratio $R_2/R_1 = 10$

Negative threshold:

$$-V_T = \frac{-V_{\text{out}}}{R_2/R_1} = \frac{-15}{10} = -1.5 \text{ V}$$

Positive threshold:

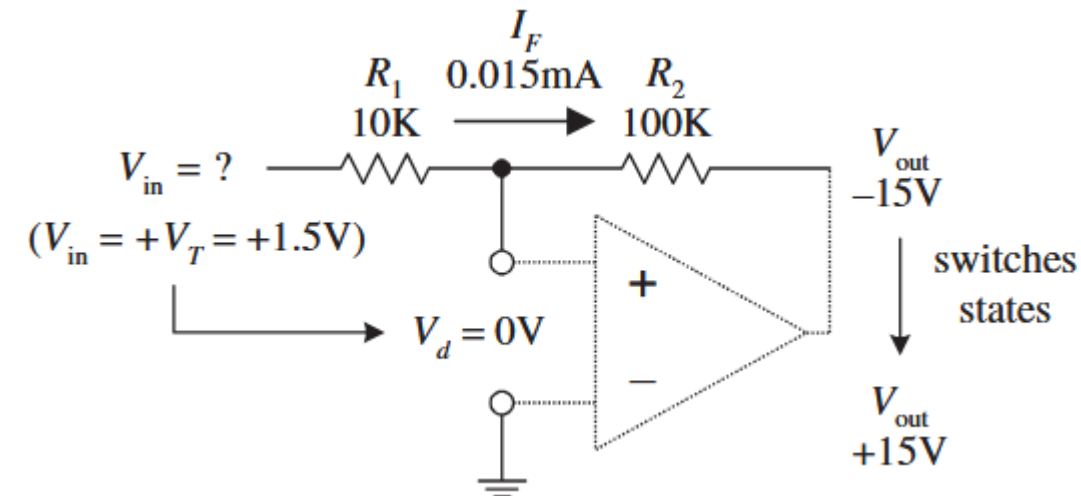
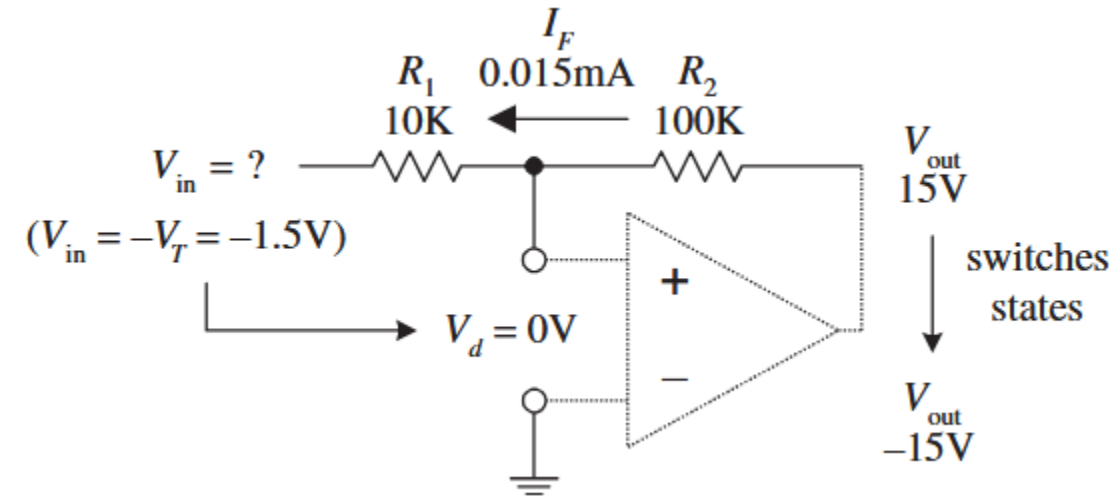
$$+V_T = \frac{+15}{10} = +1.5 \text{ V}$$

Hysteresis Voltage:

$$V_h = +V_T - (-V_T) = 3 \text{ V}$$

$$I_F = (V_{\text{out}} - V_{\text{in}}) / (R_1 + R_2)$$

$$V_d = I_F R_1$$



Type of Op-Amp

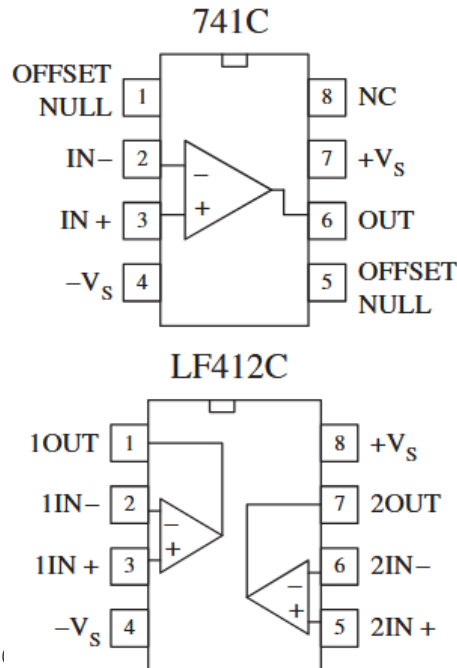
Types of Op-Amps:

General-Purpose vs Precision Op-Amps

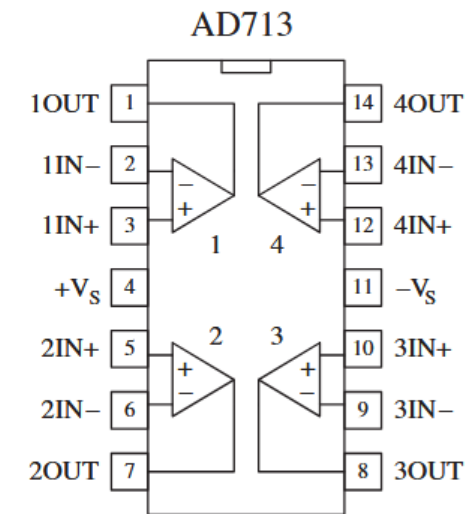
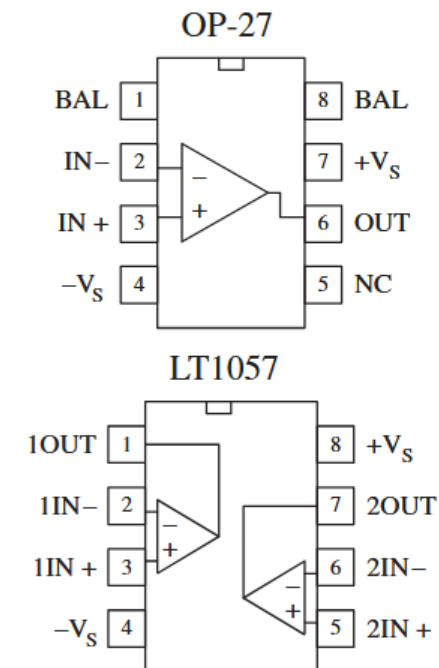
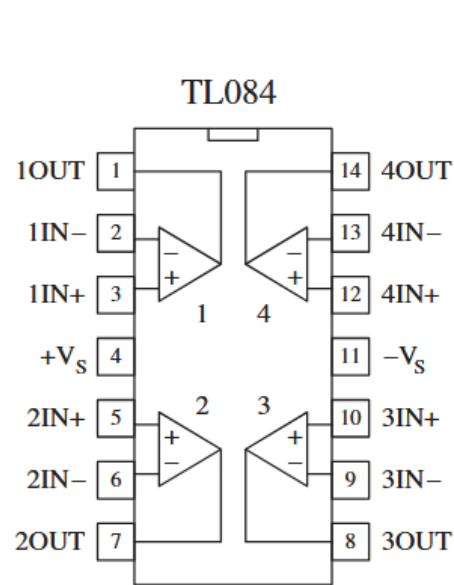
General-Purpose: Standard use in basic analog circuits.

Precision Op-Amps: Designed for low offset voltage, low bias current, and minimal drift—ideal for sensitive applications.

General Purpose



Precision



Types of Op-Amps: Technology

Bipolar Op-Amps (e.g., 741): Higher input bias current (nanoamps). Better offset voltage and thermal stability. Requires compensation resistor to minimize offset errors.

JFET Op-Amps: Very low input bias current (picoamps). Susceptible to **phase inversion** if input voltage nears the negative rail.

MOSFET Op-Amps: Extremely low input bias current (fractions of a picoamp). Medium to high offset voltage.

Hybrid Types: Combine benefits of technologies (e.g., BiFET = bipolar + JFET).

Types of Op-Amps: Technology Cont.

Feature	Bipolar	JFET	MOSFET
Offset Voltage	Low	Medium	Medium–High
Offset Drift	Low	Medium	Medium
Input Bias Current	High (nA)	Low (pA)	Very Low (sub-pA)
Bias Matching	Excellent	Fair	Fair
Temp. Variation	Low	Fair	Fair

Types of Op-Amps:

Programmable Op-Amps

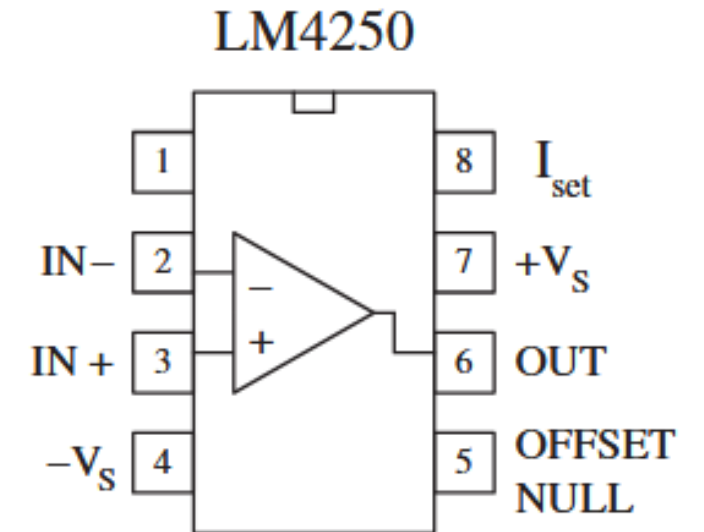
Example: LM4250

Used in low-power applications (e.g., battery-powered devices).

Characteristics such as slew rate, offset, gain-bandwidth, and noise can be set using a programming current.

Operates with low supply voltage (as low as **1 V**).

Programming current drawn via a resistor (e.g., from pin 8 to ground).



Types of Op-Amps:

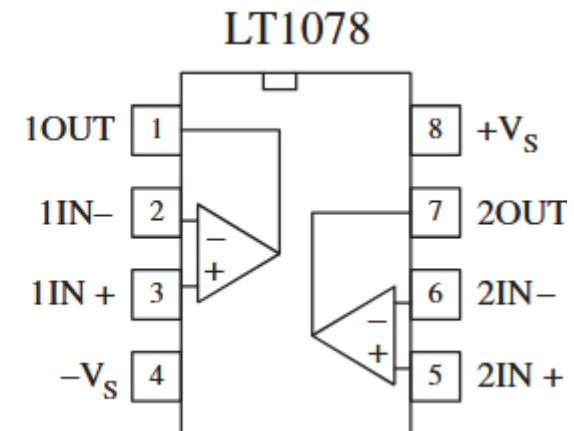
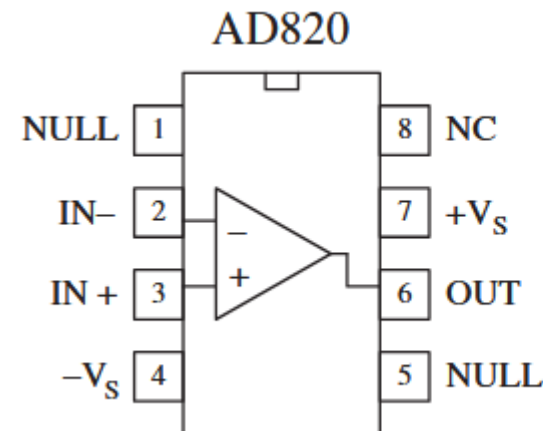
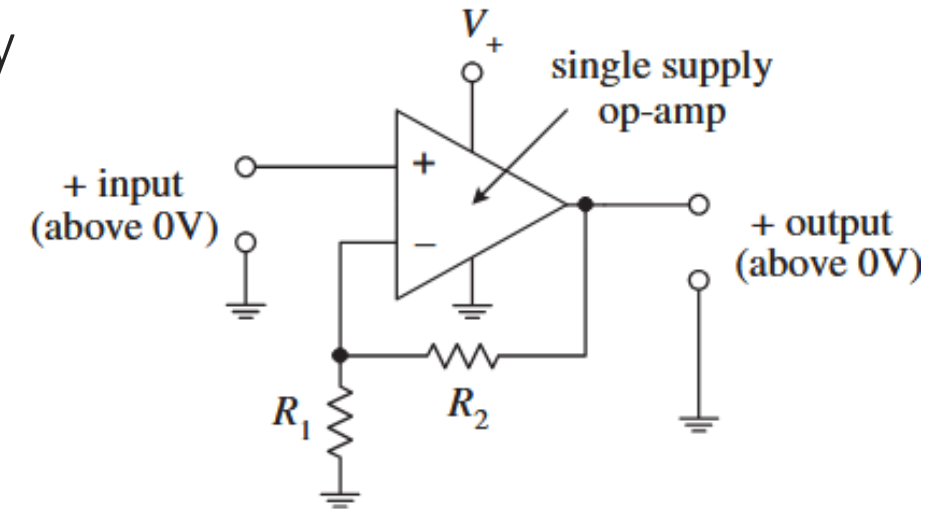
Single-Supply Op-Amps

Designed to operate with a single positive voltage supply (e.g., +12 V).

Input range includes ground, but output cannot go negative.

Suitable for **battery-powered and portable circuits**.

Not ideal for AC-coupled applications requiring bipolar swing.



Types of Op-Amps:

Audio Op-Amps

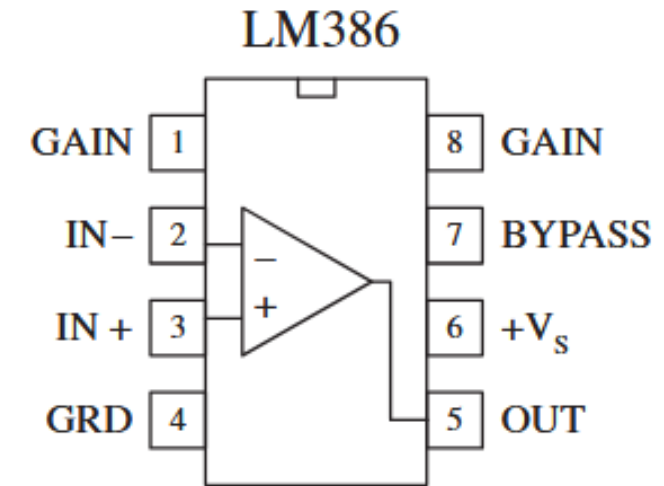
Optimized for **audio frequency range (20 Hz – 20 kHz)**.

Applications: audio preamps, radios, intercoms, automotive audio.

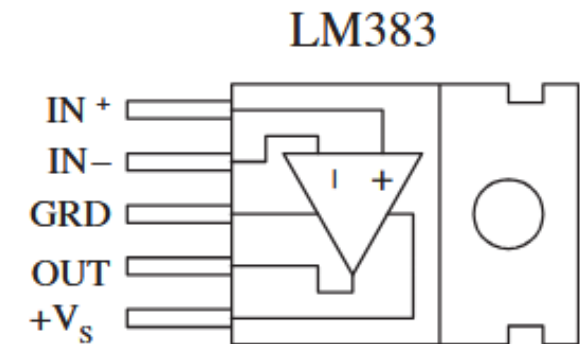
E.g.,

LM386: Low-voltage amp, gain = 20 (internally), expandable to 200. Drives 8 Ω speakers, runs on +4 to +12 V.

LM383: High-power amp, 3.5 A output current, drives 4 Ω load, includes thermal protection and heatsink.



low-voltage power amplifier



8-watt power amplifier

Op-Amp Specifications

LM741 (abt. Rs. 30/=)

The LM741 series are general purpose operational amplifiers which feature improved performance over industry standards like the LM709. They are direct, plug-in replacements for the 709C, LM201, MC1439 and 748 in most applications.

The amplifiers offer many features which make their application nearly foolproof: overload protection on the input and output, no latch-up when the common mode range is exceeded, as well as freedom from oscillations.

The LM741C is identical to the LM741/LM741A except that the LM741C has their performance ensured over a 0°C to +70°C temperature range, instead of -55°C to +125°C.

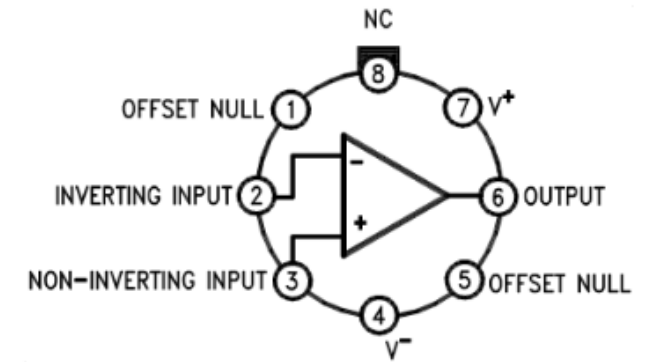


Figure 1. TO-99 Package
See Package Number LMC0008C

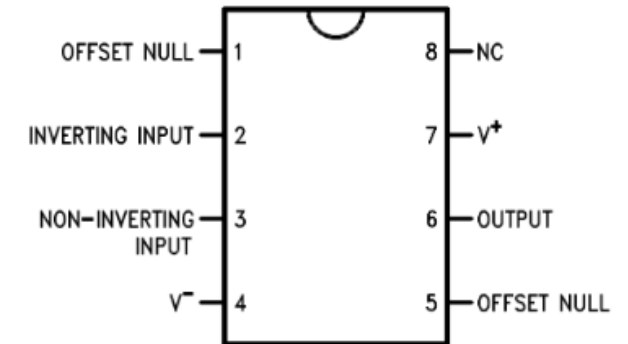


Figure 2. CDIP or PDIP Package
See Package Number NAB0008A, P0008E

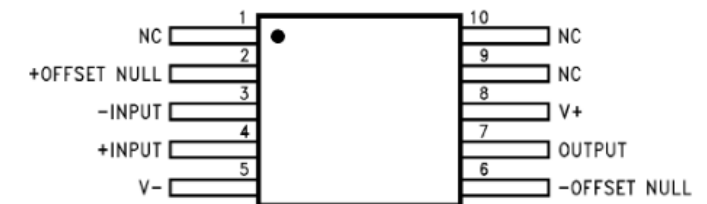


Figure 3. CLGA Package
See Package Number NAD0010A

LM741 Cont.



LM741 Pinout Diagram

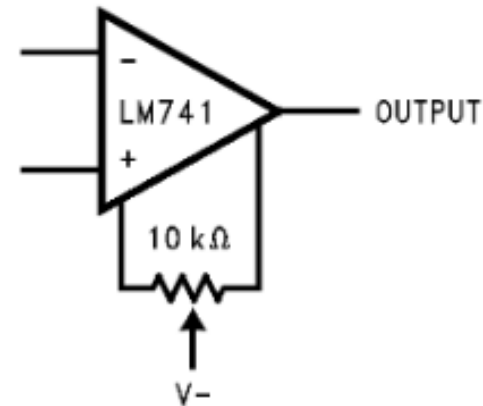
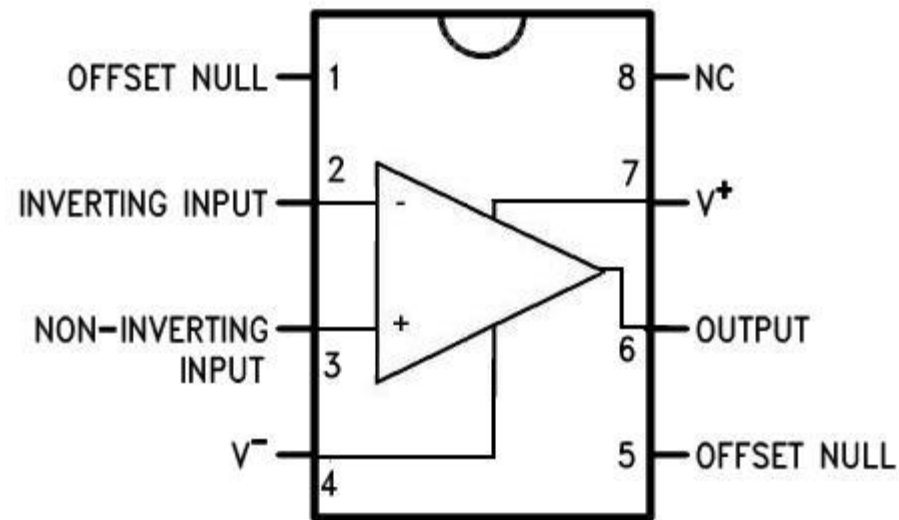


Figure 4. Offset Nulling Circuit

These devices have limited ESD protection; short leads or use conductive foam during storage or handling to prevent MOS gate damage.

Absolute Maximum Ratings⁽¹⁾⁽²⁾⁽³⁾

	LM741A	LM741	LM741C
Supply Voltage	±22V	±22V	±18V
Power Dissipation ⁽⁴⁾	500 mW	500 mW	500 mW
Differential Input Voltage	±30V	±30V	±30V
Input Voltage ⁽⁵⁾	±15V	±15V	±15V
Output Short Circuit Duration	Continuous	Continuous	Continuous
Operating Temperature Range	−55°C to +125°C	−55°C to +125°C	0°C to +70°C
Storage Temperature Range	−65°C to +150°C	−65°C to +150°C	−65°C to +150°C
Junction Temperature	150°C	150°C	100°C
Soldering Information			
P0008E-Package (10 seconds)	260°C	260°C	260°C
NAB0008A- or LMC0008C-Package (10	Differential Input Voltage Range: Max allowable voltage difference between inputs without distortion.		300°C
M-Package			
Vapor Phase (60 seconds)	215°C	215°C	215°C
Infrared (15 seconds)	215°C	215°C	215°C
ESD Tolerance ⁽⁶⁾	400V	400V	400V

- (1) “Absolute Maximum Ratings” indicate limits beyond which damage to the device may occur. Operating Ratings indicate conditions for which the device is functional, but do not ensure specific performance limits.
- (2) For military specifications see RETS741X for LM741 and RETS741AX for LM741A.
- (3) If Military/Aerospace specified devices are required, please contact the TI Sales Office/Distributors for availability and specifications.
- (4) For operation at elevated temperatures, these devices must be derated based on thermal resistance, and T_j max. (listed under “Absolute Maximum Ratings”). $T_j = T_A + (\theta_{jA} P_D)$.
- (5) For supply voltages less than ±15V, the absolute maximum input voltage is equal to the supply voltage.
- (6) Human body model, 1.5 kΩ in series with 100 pF.

Electrical Characteristics⁽¹⁾

Parameter	Test Conditions	LM741A			LM741			LM741C			Units
		Min	Typ	Max	Min	Typ	Max	Min	Typ	Max	
Input Offset Voltage	$T_A = 25^{\circ}\text{C}$ $R_S \leq 10\text{ k}\Omega$					1.0	5.0		2.0	6.0	mV
	$R_S \leq 50\Omega$		0.8	3.0							
	$T_{AMIN} \leq T_A \leq T_{AMAX}$ $R_S \leq 50\Omega$ $R_S \leq 10\text{ k}\Omega$			4.0			6.0			7.5	mV
Average Input Offset Voltage Drift				15							$\mu\text{V}/^{\circ}\text{C}$

(1) Unless otherwise specified, these specifications apply for $V_S = \pm 15\text{V}$, $-55^{\circ}\text{C} \leq T_A \leq +125^{\circ}\text{C}$ (LM741/LM741A). For the LM741C/LM741E, these specifications are limited to $0^{\circ}\text{C} \leq T_A \leq +70^{\circ}\text{C}$.

Input Offset Voltage: Small voltage that must be applied to null the output when both inputs are zero.

Electrical Characteristics⁽¹⁾ (continued)

Parameter	Test Conditions	LM741A			LM741			LM741C			Units
		Min	Typ	Max	Min	Typ	Max	Min	Typ	Max	
Input Offset Voltage Adjustment Range	$T_A = 25^\circ\text{C}$, $V_S = \pm 20\text{V}$	± 10				± 15			± 15		mV
Input Offset Current	$T_A = 25^\circ\text{C}$		3.0	30		20	200		20	200	nA
	$T_{AMIN} \leq T_A \leq T_{AMAX}$			70		85	500			300	
Average Input Offset Current Drift				0.5							nA/ $^\circ\text{C}$
Input Bias Current	$T_A = 25^\circ\text{C}$		30	80		80	500		80	500	nA
	$T_{AMIN} \leq T_A \leq T_{AMAX}$			0.210			1.5			0.8	μA
Input Resistance	$T_A = 25^\circ\text{C}$, $V_S = \pm 20\text{V}$	1.0	6.0		0.3	2.0		0.3	2.0		M Ω
	$T_{AMIN} \leq T_A \leq T_{AMAX}$, $V_S = \pm 20\text{V}$	0.5									

Input Offset Current: Difference in bias currents between the two inputs. Caused by mismatch in internal input circuitry. Can be corrected using offset adjustment pins or a potentiometer.

Input Bias Current: Average current drawn by both inputs (due to finite input impedance). Causes voltage drops in external resistors → can affect output. **FET op-amps:** Low bias current (pA); **Bipolar op-amps:** Higher (nA).

Parameter	Test Conditions	LM741A			LM741			LM741C			Units
		Min	Typ	Max	Min	Typ	Max	Min	Typ	Max	
Large Signal Voltage Gain	$T_A = 25^\circ\text{C}$, $R_L \geq 2\text{ k}\Omega$ $V_S = \pm 20\text{V}$, $V_O = \pm 15\text{V}$ $V_S = \pm 15\text{V}$, $V_O = \pm 10\text{V}$	50			50	200		20	200		V/mV
	$T_{\text{AMIN}} \leq T_A \leq T_{\text{AMAX}}$, $R_L \geq 2\text{ k}\Omega$, $V_S = \pm 20\text{V}$, $V_O = \pm 15\text{V}$ $V_S = \pm 15\text{V}$, $V_O = \pm 10\text{V}$ $V_S = \pm 5\text{V}$, $V_O = \pm 2\text{V}$	32			25			15			V/mV
		10									
Output Voltage Swing	$V_S = \pm 20\text{V}$ $R_L \geq 10\text{ k}\Omega$ $R_L \geq 2\text{ k}\Omega$	± 16 ± 15									V
	$V_S = \pm 15\text{V}$ $R_L \geq 10\text{ k}\Omega$ $R_L \geq 2\text{ k}\Omega$				± 12 ± 10	± 14 ± 13		± 12 ± 10	± 14 ± 13		V

Voltage Gain (AV): High DC gain: Typically, 10^4 to 10^6 (80–120 dB). Drops with frequency; unity gain at f_T (typically 1–10 MHz).

Output Voltage Swing: Max peak output (relative to ground or supply rails) without clipping.

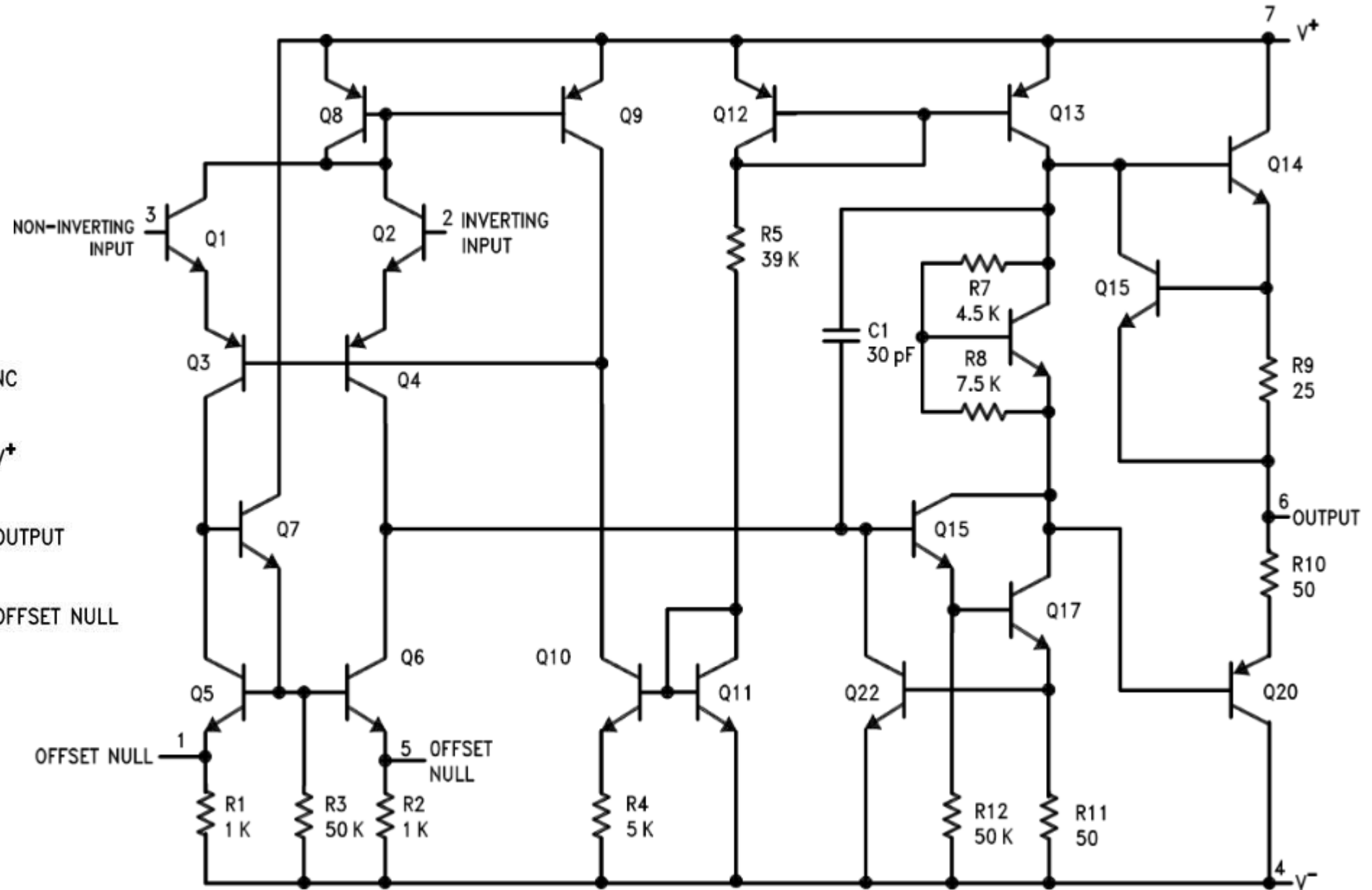
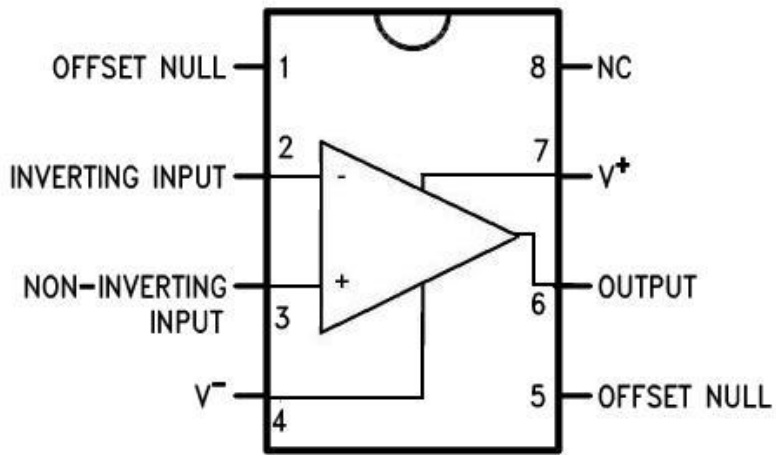
Parameter	Test Conditions	LM741A			LM741			LM741C			Units
		Min	Typ	Max	Min	Typ	Max	Min	Typ	Max	
Output Short Circuit Current	$T_A = 25^\circ\text{C}$ $T_{AMIN} \leq T_A \leq T_{AMAX}$	10 10	25	35 40		25			25		mA
Common-Mode Rejection Ratio	$T_{AMIN} \leq T_A \leq T_{AMAX}$ $R_S \leq 10\text{ k}\Omega, V_{CM} = \pm 12\text{V}$ $R_S \leq 50\Omega, V_{CM} = \pm 12\text{V}$				70	90		70	90		dB
Supply Voltage Rejection Ratio	$T_{AMIN} \leq T_A \leq T_{AMAX},$										dB
Transient Response Rise Time Overshoot			6.0	20		5			5		
Bandwidth ⁽²⁾	$T_A = 25^\circ\text{C}$	0.437	1.5								MHz
Slew Rate	$T_A = 25^\circ\text{C}$, Unity Gain	0.3	0.7			0.5			0.5		V/ μs
Supply Current	$T_A = 25^\circ\text{C}$					1.7	2.8		1.7	2.8	mA
Power Consumption	$T_A = 25^\circ\text{C}$ $V_S = \pm 20\text{V}$ $V_S = \pm 15\text{V}$		80	150		50	85		50	85	mW

Supply Current: Quiescent current drawn from the power supply (no load, zero output).

Common-Mode Rejection Ratio (CMRR): Measures how well an op-amp rejects common-mode signals (same signal at both inputs). $\text{CMRR} = \text{Differential Gain} / \text{Common-Mode Gain}$. Higher CMRR $\rightarrow$ Better ability to amplify only the difference between inputs.

LM741

LM741 Pinout Diagram



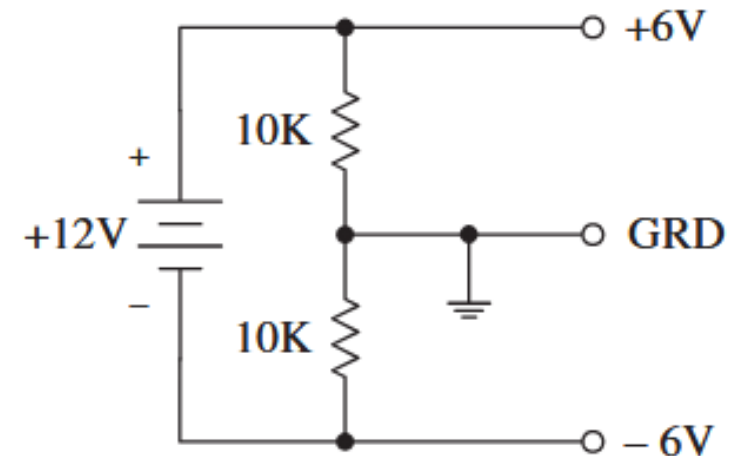
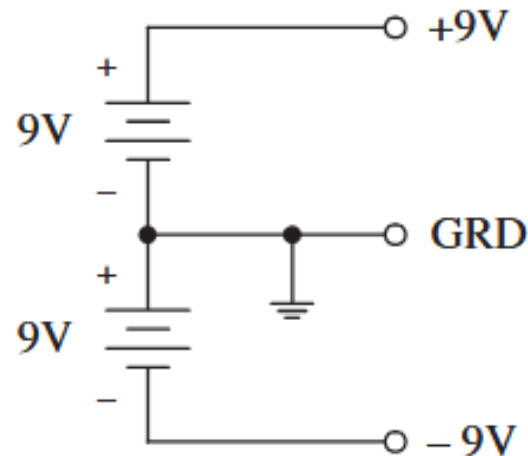
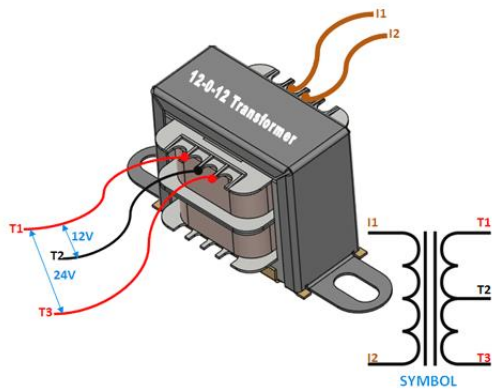
Comparison

TYPE	TOTAL SUPPLY VOLTAGE		SUPPLY CURRENT (mA)	OFFSET VOLTAGE		CURRENT		SLEW RATE TYPICAL (V/ μ S)	f_T TYPICAL (MHz)	CMRR MIN (dB)	GAIN MIN (mA)	OUTPUT CURRENT MAX (mA)
	MIN (V)	MAX (V)		TYPICAL (mV)	MAX (mV)	BIAS MAX (nA)	OFFSET MAX (nA)					
Bipolar 741C	10	36	2.8	2	6	500	200	0.5	1.2	70	86	20
MOSFET CA3420A	2	22	1	2	5	0.005	0.004	0.5	0.5	60	86	2
JFET LF411	10	36	3.4	0.8	2	0.2	0.1	15	4	70	88	30
Bipolar, precision LM10	1	45	0.4	0.3	2	20	0.7	0.12	0.1	93	102	20

Powering Op-Amps: Dual vs Single Supply

Dual-Supply Operation ($\pm V$): Most op-amp applications use $\pm 15\text{ V}$ dual-polarity supplies. Provides symmetrical swing around 0 V (ideal for AC signals). Can be implemented using a **center-tapped transformer**.

Battery-Powered and Single-Supply Operation: Dual supplies are **impractical** in small, portable, or battery-powered circuits. **Single-supply op-amps** can be used, but: output **clips** when the input goes negative (below ground) and **unsuitable** for AC-coupled signals without modification.



Powering Op-Amps: Dual vs Single Supply

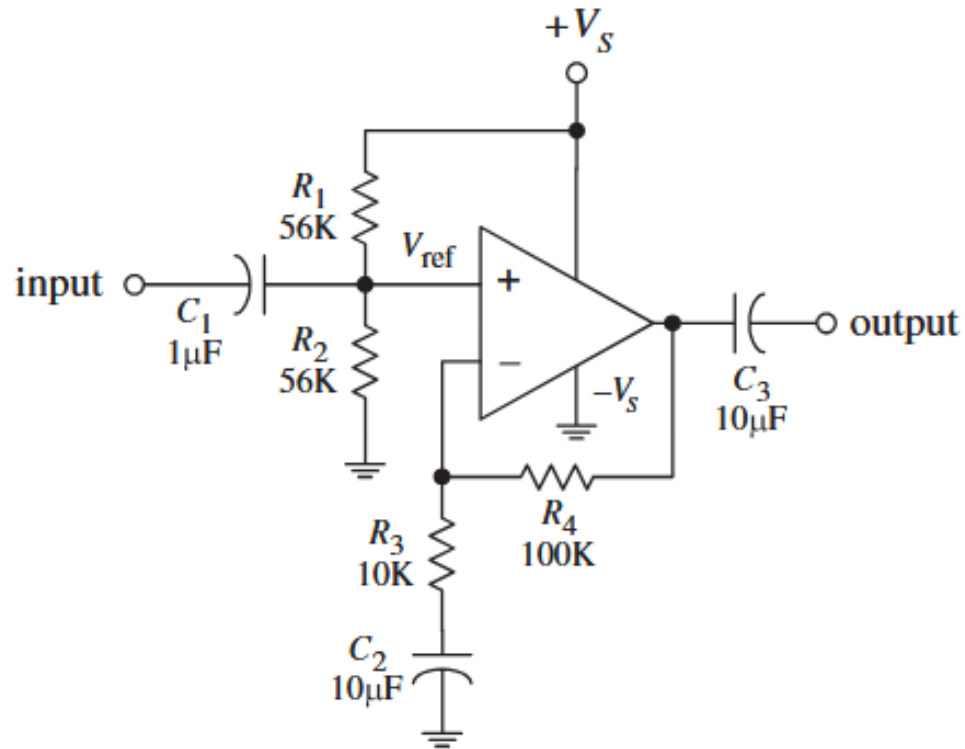
Cont.

Biasing Conventional Op-Amps on a Single Supply: Add a **DC bias** (via voltage divider) to shift signal range above ground. Allows conventional op-amp to operate without input/output clipping. Bias voltage (often $\frac{1}{2} V_{CC}$) creates a **virtual ground**. Input/output capacitors **block DC** and **couple AC signals**.

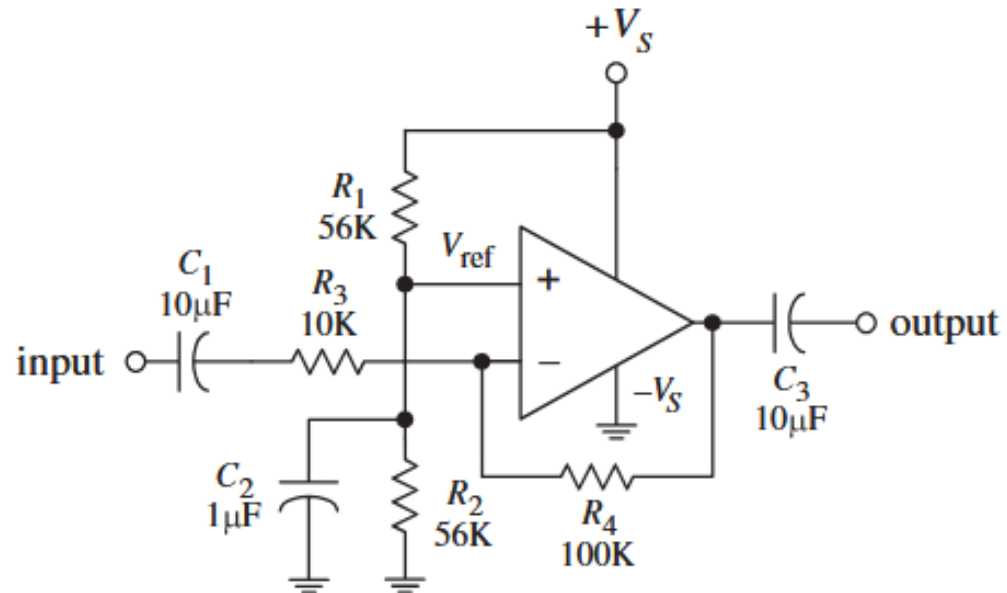
AC-Coupled Op-Amp Circuits (Single Supply): Bias network (R1, R2) sets input at $\frac{1}{2} V_{CC}$. **C1**: Input coupling capacitor → blocks DC. **C3**: Output coupling capacitor → blocks output DC offset.

Powering Op-Amps: Dual vs Single Supply Cont.

Noninverting single-supply ac amplifier



Inverting single-supply ac amplifier



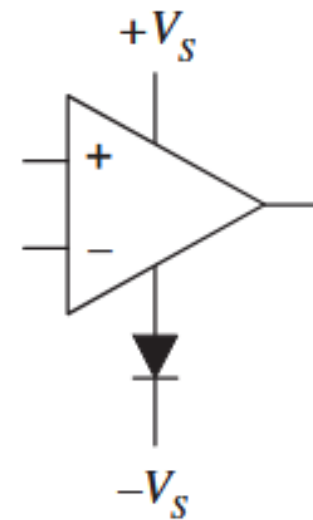
$$C_1 = \frac{1}{2\pi f_{3dB} R_1}$$
$$C_3 = \frac{1}{2\pi f_{3dB} R_{\text{load}}}$$

Further Design Guidelines

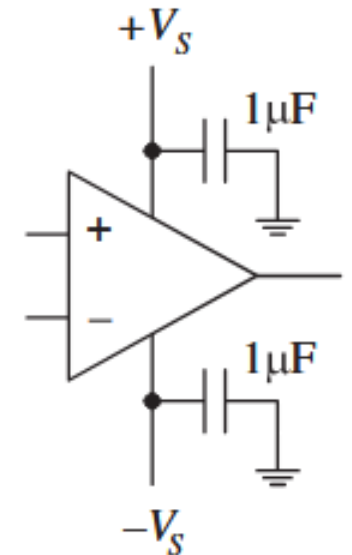
Power Supply Polarity: Never reverse op-amp supply leads – it can **permanently damage** the IC. **Protection Tip:** Add a **diode** (e.g., 1N4007) between the op-amp's **negative supply terminal** and the power source to prevent reverse polarity damage.

Supply Wiring Best Practices: Keep **supply wires short and direct** to reduce noise and unwanted oscillations. Use **bypass capacitors** from supply pins to ground (**0.1 μF ceramic** or **1.0 μF tantalum**) to stabilize supply voltage and suppress transient disturbances.

Reverse-polarity protection



Oscillation prevention



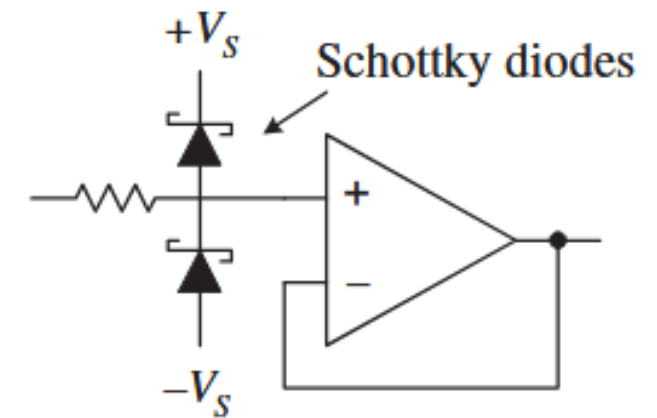
Further Design Guidelines Cont.

Avoiding Latch-Up Conditions:

Inputs must stay **within the supply voltage range**: Do not exceed $+V_S + 0.7\text{ V}$ or $-V_S - 0.7\text{ V}$. Exceeding limits can cause **internal short circuits** $\rightarrow$ IC destruction.

Critical Warning: If a signal is present **before power is applied**, the op-amp may **latch up or fail** immediately.

Destructive latch-up protection



Input Protection Techniques:

Clamp inputs with Schottky diodes to prevent overvoltage (fast, low forward-voltage diodes are preferred).

Add **series current-limiting resistors** to control inrush current through the diodes.

Note: Diodes may introduce **leakage currents**, affecting precision in high-accuracy circuits.

Voltage and Current Offset Compensation

Ideal case: Output = 0 V when both inputs = 0 V.

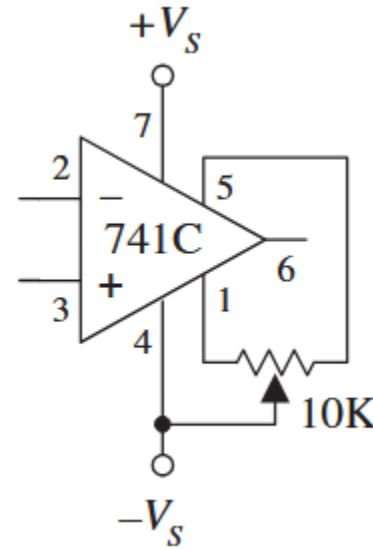
Reality: Internal mismatches → small unwanted output (μV to mV range).

Input Offset Voltage: Voltage that must be applied to one input to null the output.

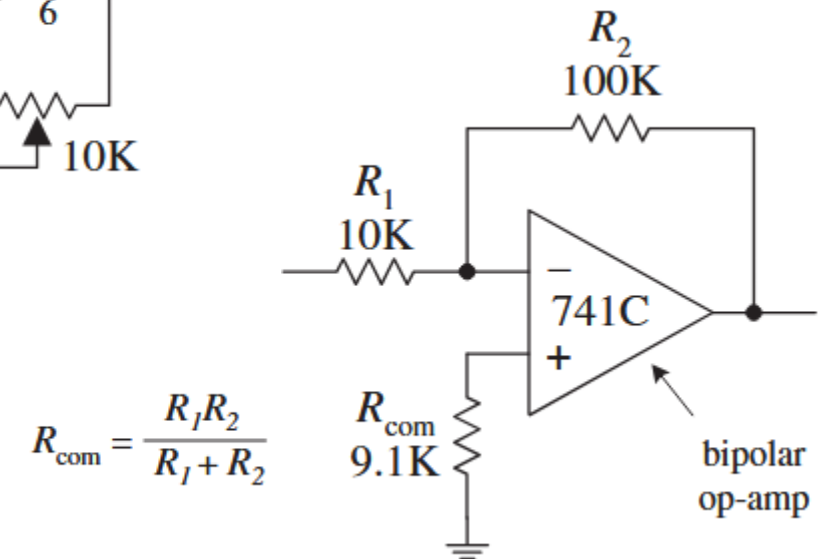
Many op-amps include **offset null terminals**.

Compensation Resistor: Place a resistor between **non-inverting input** and ground.

Input offset voltage adjustment



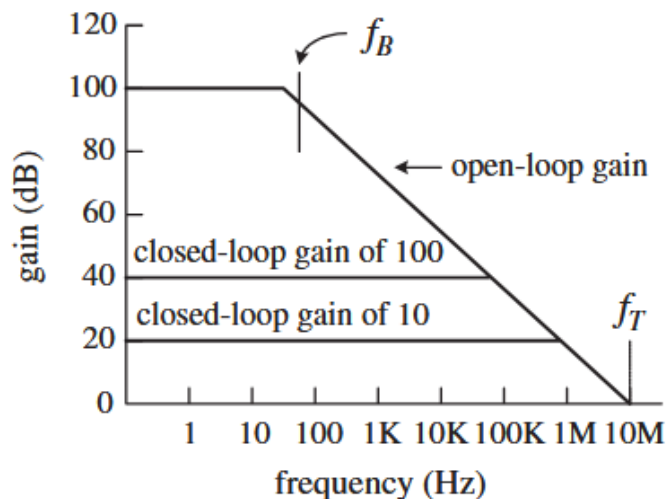
Input offset current adjustment



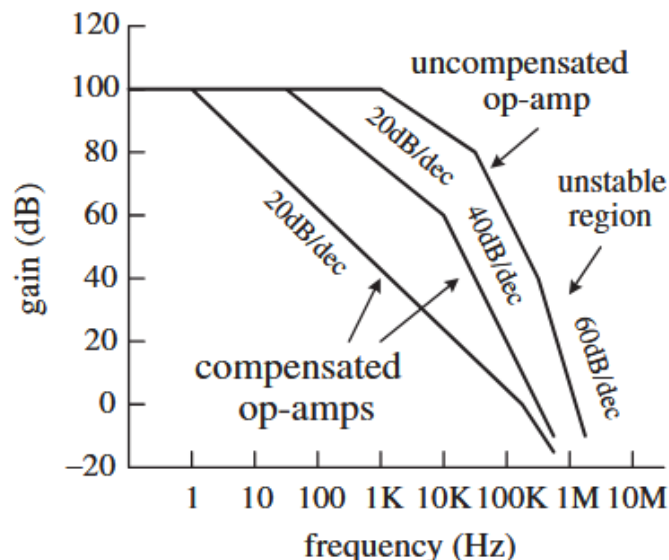
Frequency Compensation

A typical op amp has a high open-loop gain (10^4 – 10^6), but this decreases with frequency due to internal low-pass behavior. Gain falls to 1 at the unity-gain frequency (~ 1 MHz). To prevent instability and oscillations from phase shifts, use frequency compensation—or choose an internally compensated op amp.

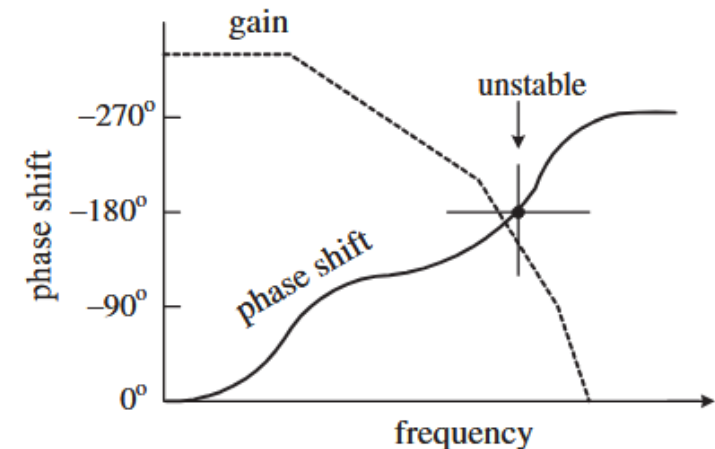
Open-loop and closed-loop responses for a typical op amp



Compensated and uncompensated op amp response curves



Internal phase shifts



Comparators

A **comparator** detects whether one voltage is **greater than another**.

Common use: **signal threshold detection** (e.g., zero-crossing, level detection).

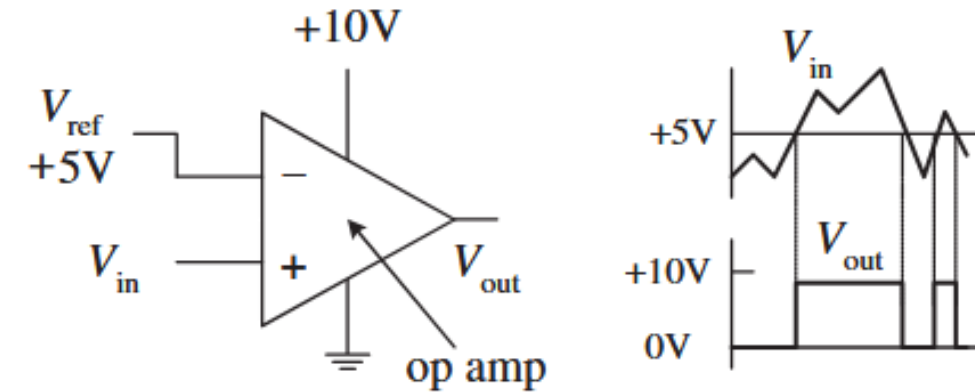
Noninverting Comparator: Output switches **low** → **high** when input > reference (at inverting input).

Inverting Comparator: Output switches **high** → **low** when input > reference (at noninverting input).

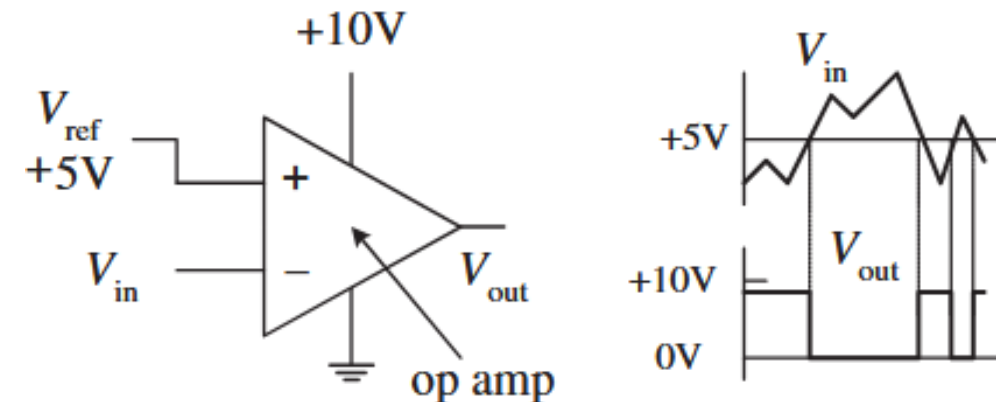
Reference voltage can be set using a **voltage divider (potentiometer)**.

Note: Not all op-amps work well near ground ($V_- = 0\text{ V}$); use caution.

Noninverting comparator

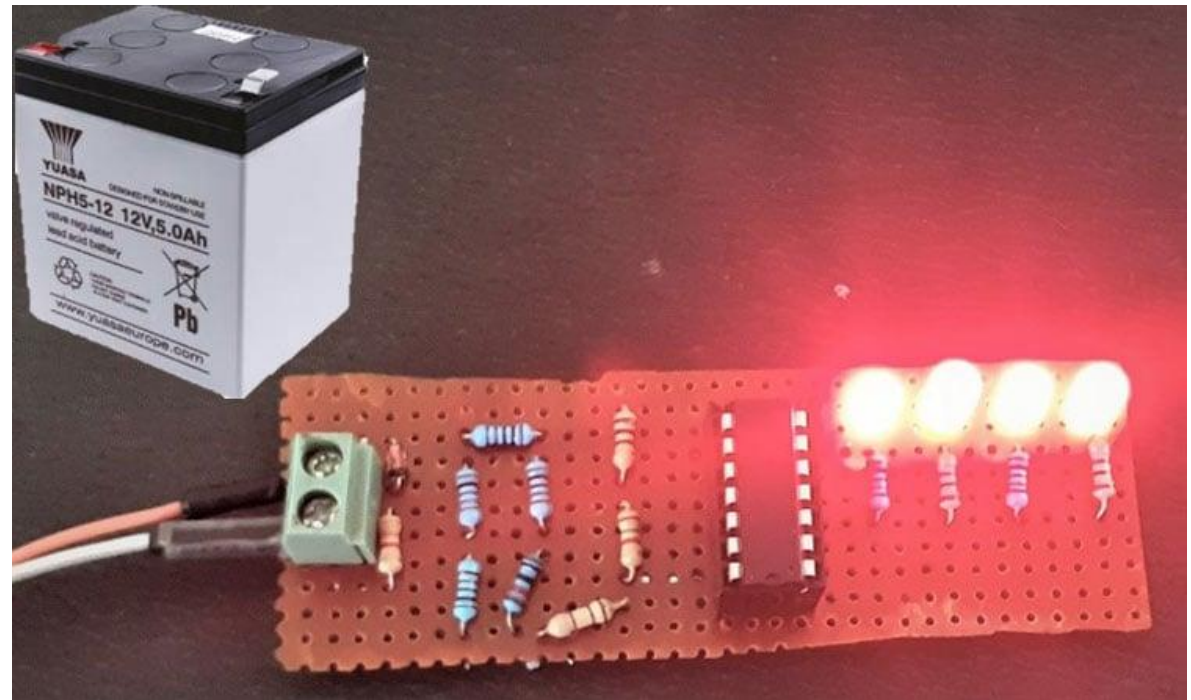
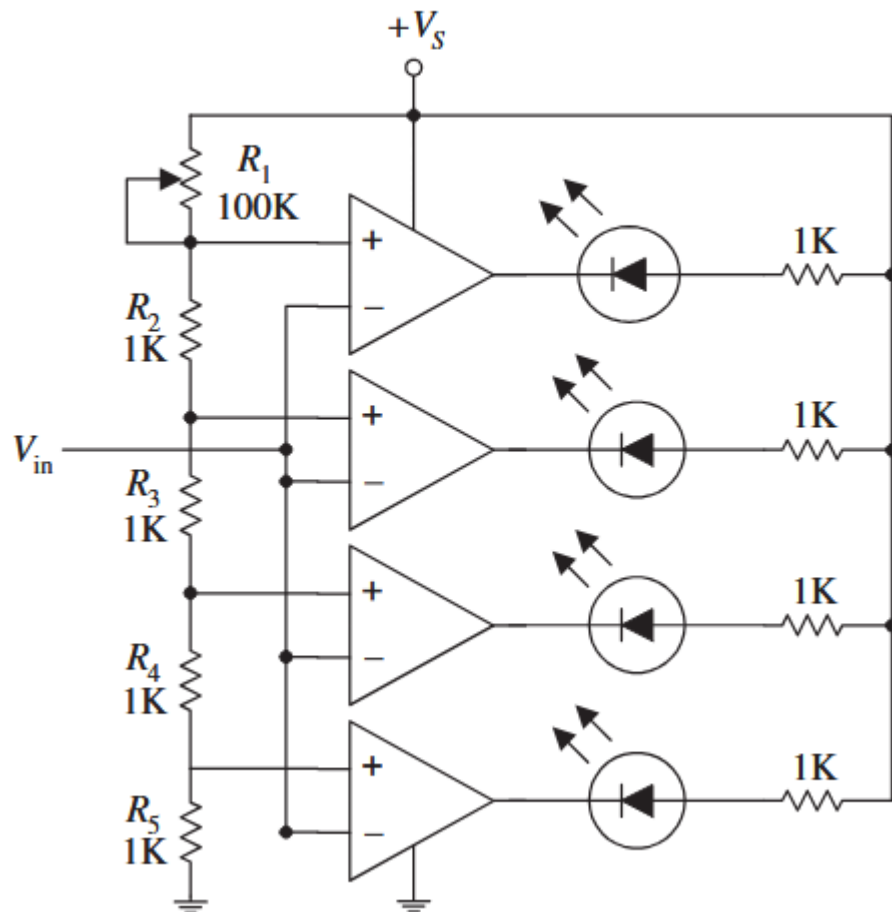


Inverting comparator



Voltage-Level Indicator

LED voltage-level display

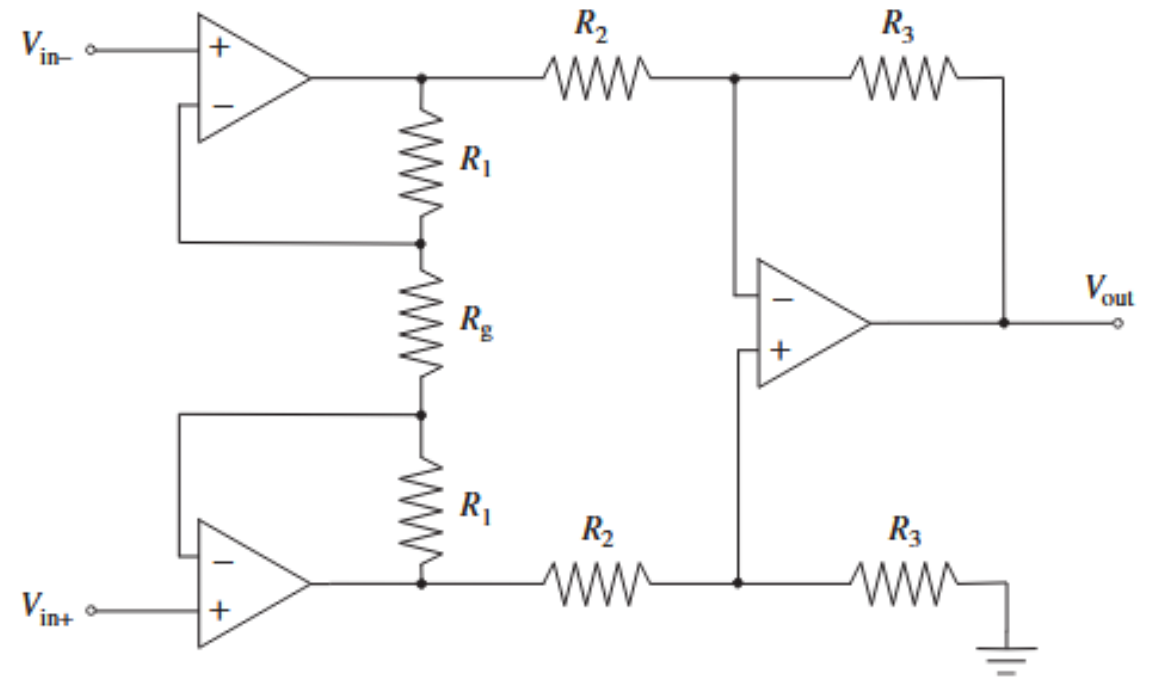


Instrumentation Amplifiers

Amplify the **difference** between two input signals and suppress common-mode signals (e.g., noise).

E.g., **ECG (Electrocardiogram)**: Detects tiny voltage differences between body electrodes to monitor heart activity.

Implementation: Can be built using **three op-amps** and matched resistors. More commonly used as **dedicated ICs** (e.g., AD620, INA128) with factory-trimmed resistor networks.

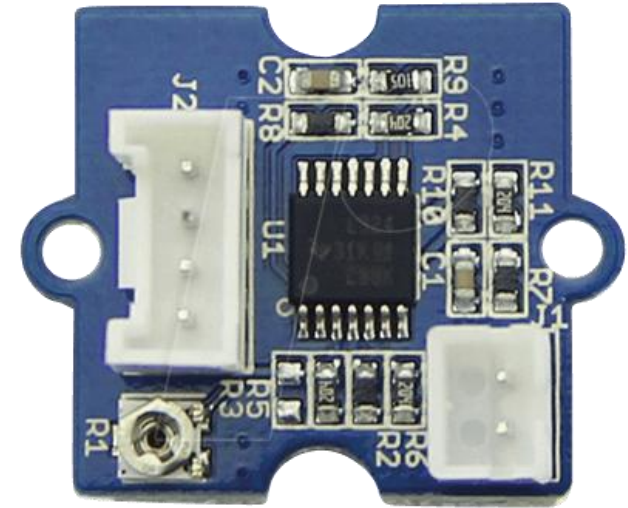
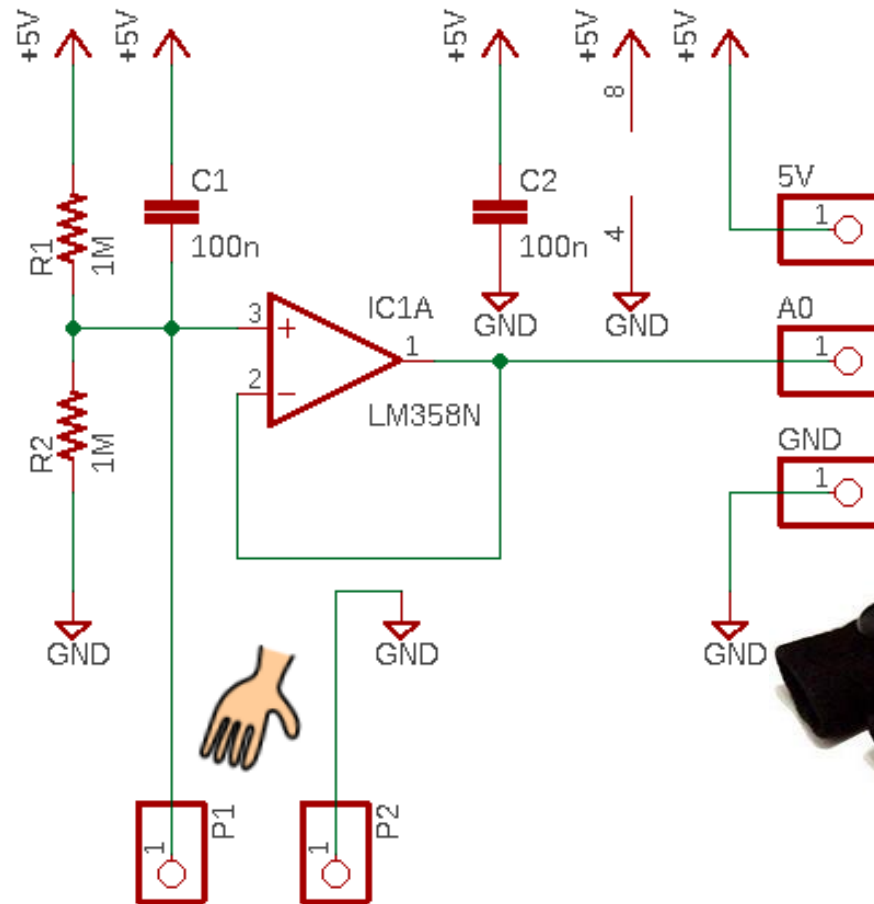


$$\frac{V_{out}}{V_{in+} - V_{in-}} = \left(1 + \frac{2R_1}{R_g} \right) \frac{R_3}{R_2}$$

Case Study: Grove GSR Sensor

https://wiki.seeedstudio.com/Grove-GSR_Sensor/

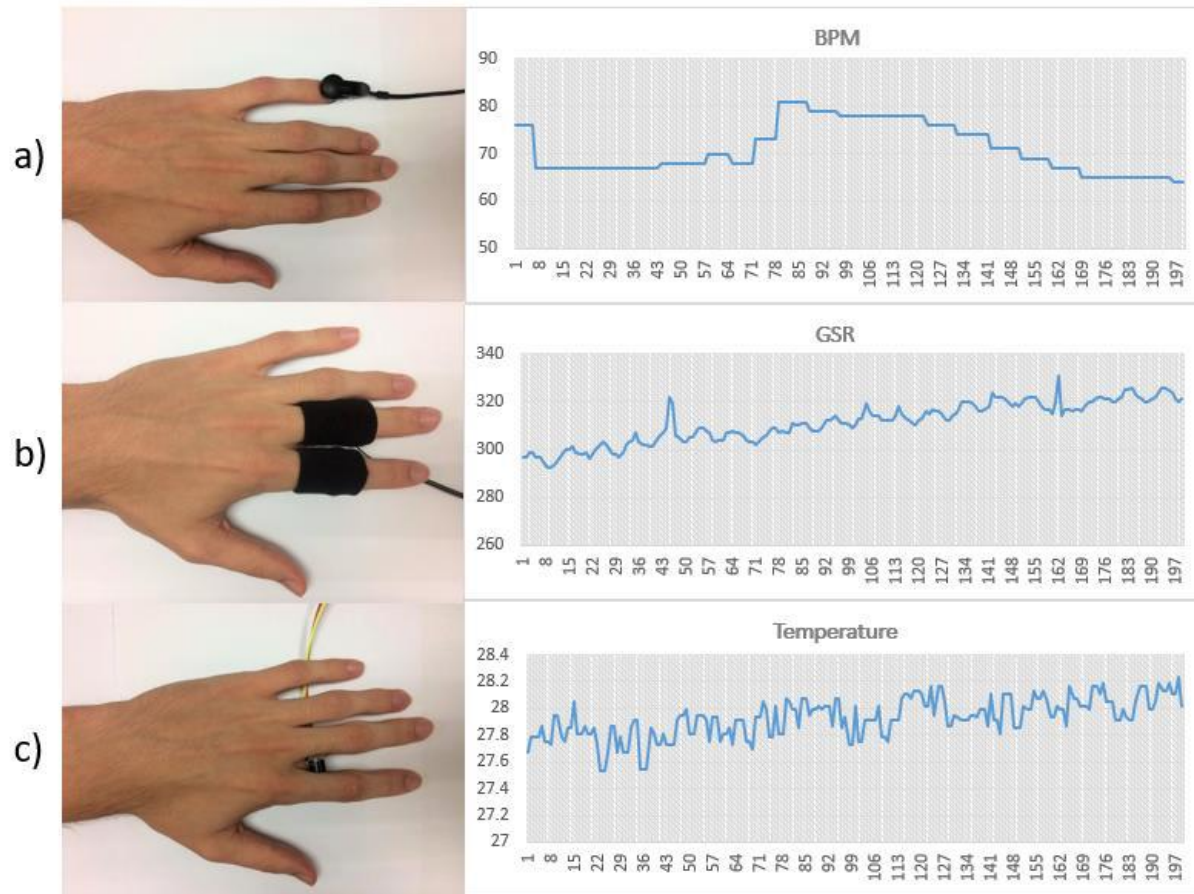
A bio-signal sensor used to measure the **galvanic skin response (GSR)**—also known as **electrodermal activity (EDA)**. It indicates how the electrical conductance of the skin changes due to moisture (sweat), which varies with psychological or physiological arousal (e.g., stress, excitement).



Galvanic Skin Response (GSR/EDA)

Sensors positions

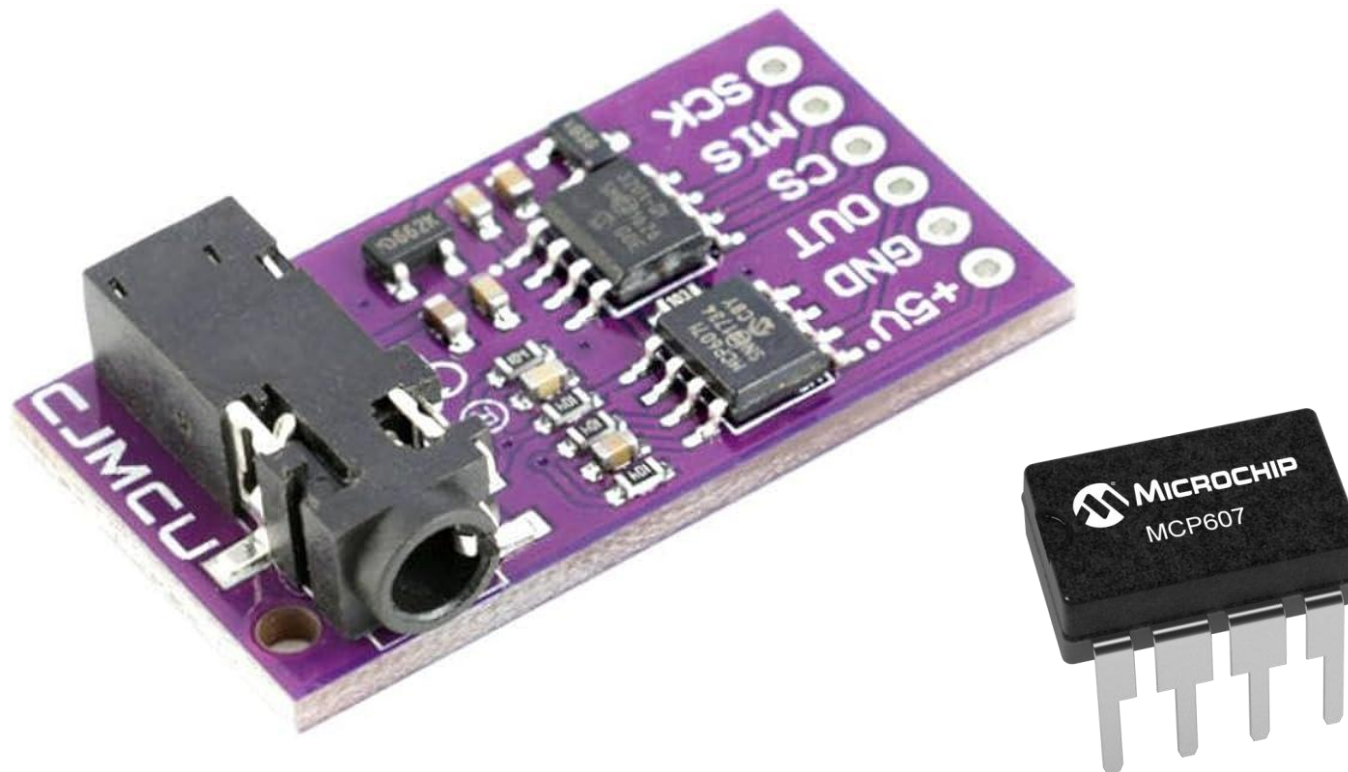
Waveform



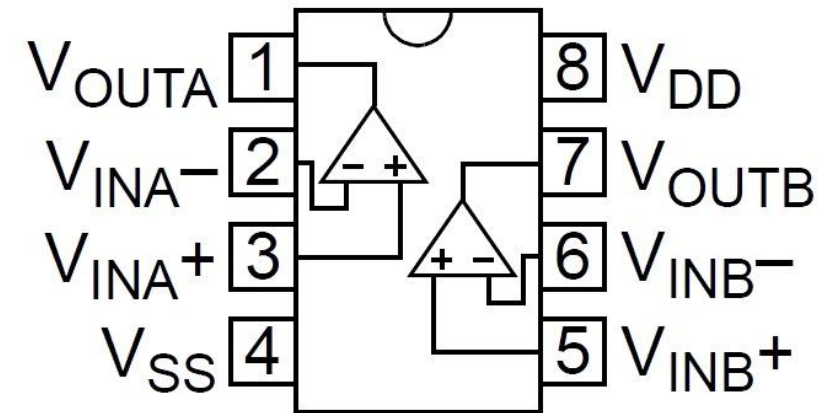
CJMCU-6701 GSR Skin Sensor Module

Analog SPI 3.3V/5V for Arduino

MCP607 2.5-6.0V CMOS Op Amp, **MCP3201** 2.7V 12-Bit A/D Converter with SPI Serial Interface, MCP1541 2.5V, and 4.096V



MCP607
PDIP, SOIC, TSSOP



Emotiv EPOC / BCI



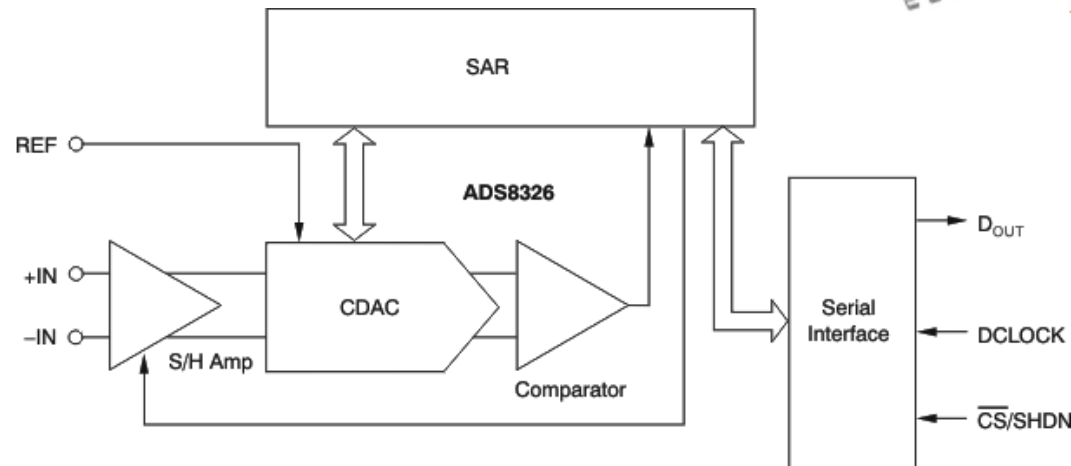
The ADS8326 is a 16-bit, sampling, analog-to-digital (A/D) converter specified for a supply voltage range from 2.7V to 5.5V. It requires very little power, even when operating at the full data rate. At lower data rates, the high speed of the device enables it to spend most of its time in the power-down mode. For example, the average power dissipation is less than 0.2mW at a 10kHz data rate.



PRECISION ADCS ($\leq 10\text{MSPS}$)

ADS8326 ✓ ACTIVE

16-Bit, Pseudo-Diff Input, 250kSPS Serial Out, 2.7V-to-5.5V Micropower Sampling ADC



APPLICATIONS

- Battery-Operated Systems
- Remote Data Acquisition
- Isolated Data Acquisition
- Simultaneous Sampling, Multichannel Systems
- Industrial Controls
- Robotics
- Vibration Analysis



Thanks